



PAPER • OPEN ACCESS

Empirically-calibrated H100 node power models for accurate AI training energy estimation

To cite this article: Alex C Newkirk *et al* 2025 *Environ. Res.: Energy* **2** 045016

View the [article online](#) for updates and enhancements.

You may also like

- [How to estimate carbon footprint when training deep learning models? A guide and review](#)
Lucía Bouza, Aurélie Bugeau and Loïc Lannelongue
- [The cost of computing](#)
Mark Klimek and Keith Pearce
- [Quantum Enhanced Josephson Junction Field-Effect Transistors for Low-Energy Power-Efficient Microelectronics Applications](#)
Wei Pan

ENVIRONMENTAL RESEARCH ENERGY



PAPER

OPEN ACCESS

RECEIVED

11 August 2025

REVISED

20 November 2025

ACCEPTED FOR PUBLICATION

26 November 2025

PUBLISHED

8 December 2025

Original content from this work may be used under the terms of the [Creative Commons Attribution 4.0 licence](#).

Any further distribution of this work must maintain attribution to the author(s) and the title of the work, journal citation and DOI.



Empirically-calibrated H100 node power models for accurate AI training energy estimation

Alex C Newkirk^{1,2,*} , Jared Fernandez³, Jonathan Koomey⁴, Imran Latif⁵, Emma Strubell³, Arman Shehabi⁶ and Constantine Samaras^{2,7,8}

¹ Building Technologies and Urban Systems, Energy Technologies Area, Lawrence Berkeley National Laboratory, Berkeley, CA, United States of America

² Department of Engineering and Public Policy, Carnegie Mellon University, Pittsburgh, PA, United States of America

³ School of Computer Science, Carnegie Mellon University, Pittsburgh, PA, United States of America

⁴ Koomey Analytics, San Francisco, CA, United States of America

⁵ Computing and Data Sciences Directorate, Brookhaven National Laboratory, Upton, NY, United States of America

⁶ Energy Analysis and Environmental Impacts Division, Energy Technologies Area, Lawrence Berkeley National Laboratory, Berkeley, CA, United States of America

⁷ Department of Civil and Environmental Engineering, Carnegie Mellon University, Pittsburgh, PA, United States of America

⁸ Wilton E. Scott Institute for Energy Innovation, Carnegie Mellon University, Pittsburgh, PA, United States of America

* Author to whom any correspondence should be addressed.

E-mail: acnewkirk@lbl.gov

Keywords: AI energy use, accelerator power demand, computational infrastructure requirements

Abstract

Accurately quantifying the energy use of artificial intelligence (AI) training is critical for infrastructure planning, carbon accounting, and sustainable data center operation, but few studies have directly measured the power consumption of production workloads on contemporary hardware. By combining empirical measurements from Brookhaven National Laboratory during AI training on 8-graphics-processing-unit H100 systems with open-source benchmarking data, we develop statistical models relating computational intensity to node-level power consumption. We measure the gap between manufacturer-rated thermal design power (TDP) and actual power demand during AI training. Our analysis reveals that even computationally intensive workloads operate at only 76% of the 10.2 kW TDP rating. Our architecture-specific model, calibrated to floating-point operations, predicts energy consumption with 11.4% mean absolute percentage error, significantly outperforming TDP-based approaches (27%–37% error). We identified distinct power signatures between transformer and convolutional neural network architectures, with transformers showing characteristic fluctuations that may impact grid stability. These results provide a measurement-grounded basis for improving AI training energy estimates, enabling more reliable infrastructure sizing, cost projections, and environmental impact assessments.

1. Introduction

1.1. Background and motivation

The increased capabilities of artificial intelligence (AI) applications have led to their rapid integration into business and consumer ecosystems. This demand has been accompanied by large-scale investment in the data centers [1]. Properly anticipating the significant power and transmission infrastructure needs of future data center facilities, as well as assessing their growing environmental impact, depends on accurately measuring hardware power consumption and understanding how AI computation differs from traditional enterprise workloads.

AI can refer to any applications where analytic insights, novel images, or text are generated through computer algorithms, as well as image classification, natural language processing, and advanced automation. When we refer to AI here, we are specifically discussing the computationally intensive applications built on machine learning (ML) techniques. Innovations of the last decade such as deep-neural-nets (DNNs) [2] and the transformer architecture [3] have enabled both greater performance in AI models

and greater computational efficiency on parallel hardware. These advancements repeatedly illustrated ‘the bitter lesson’ of AI research: the most dramatic performance improvements have been seen in ‘methods that continue to scale with increased computation, even as the available computation becomes very great’ [4] including deep learning techniques. This scaling relationship between computation and capability has profound implications for facility infrastructure, as improvements in model performance have consistently been reinvested in larger models with greater computational demands.

Modern AI systems are composed from three fundamental inputs: datasets that contain information about our world, algorithms that process that information, and physical computer infrastructure to implement the data processing algorithms [5]. Training data, which can vary from physical modeling parameters to natural language to images, are vectorized [6] (converted into a column of numeric values). This vector object can then be transformed using matrix operations. This is the mechanical core of DNN AI, the weightings and matrix operations are derived to best predict an output given some input data. A relevant feature of this computation is its floating-point simplicity compared to conventional high-performance computing (HPC) workloads; while each individual operation is relatively straightforward, a tremendous quantity of these operations must be performed.

Initially the model has either no weighting or random weightings, developing the final relationships between input and output data entails feeding the training set through the model, evaluating the predictive performance, adjusting the parameters, and repeating. Once developers have built an optimized model, they can then feed it queries and receive outputs based on its internal weightings. The process of developing the model weights is known as ‘training’ the model, while querying a completed model and receiving output is known as ‘inference’. The key insight from this is that AI applications require large quantities of mathematical operations that are not individually complicated, which makes them uniquely suited to parallel computation hardware.

Transformer architecture DNNs benefit from a variety of parallelization techniques including data parallelism, pipeline parallelism (PP), and tensor parallelism (TP)⁹ [7], which unlock increased computational performance. Logic chips optimized for AI related workloads are sometimes collectively referred to as AI accelerators/accelerated processors or ‘AI Chips’ [8]. The most common type of accelerator is the graphics-processing-unit (GPU), but there are a variety of accelerator architectures. For a more detailed discussion of the AI hardware stack, see appendix A. Each parallelism technique affects hardware utilization differently: TP shards the matrix operations for an individual layer across devices, PP distributes model layers across devices, while context parallelism (CP) (and sequence parallelism) divides attention operations across GPUs to reduce memory requirements for processing long sequences [9].

The bulk of the computational performance supporting AI is delivered by AI accelerators. Groups of AI chips are typically integrated with one or more central-processing-units (CPUs) to coordinate workloads, supervise data transfer, and interface with other systems, as well as significant memory systems, and high-bandwidth networking components. This bundle of hardware is often referred to as a compute node, a term analogous to server, with a group of nodes referred to as a cluster.

Electricity use in data centers is largely driven by the servers, data storage, and networking equipment they house. Computation generates resistive heat, which must be removed from the data center to maintain equipment reliability and longevity. The total electricity use of the facility is directly related to the electricity used by the information technology (IT) equipment, which is why accurately assessing IT electricity use in these facilities is critical. Analyses of total data center electricity use have traditionally characterized the non-IT electricity use in data centers using a ratio called power usage effectiveness (PUE).

PUE is defined as the total electricity use of the facility divided by the IT electricity use of the facility measured at the plug. It is a ratio that is always larger than 1.0, and the fraction above 1.0 represents the losses and electricity consumed by external fans, pumps, power distribution, and cooling (generally characterized as ‘infrastructure electricity use’). The form of this ratio embodies the primacy of IT electricity use in the total electricity used by the facility. If IT electricity use goes up, the infrastructure electricity use goes up in direct proportion, all other things being equal. Errors made in assessing IT electricity use therefore compound across any model or measure of facility energy use.

Our analysis focuses specifically on IT electricity use for several reasons. First, operating electricity has historically dominated data center environmental footprint, accounting for over 90% of lifecycle

⁹ Tensors are mathematical objects that satisfy certain transformation relationships conceptually similar to vectors and matrices. Tensors are commonly used in physics and engineering, and computing their time evolution requires matrix operations akin to machine learning. Accordingly, the term tensor is sometimes shorthand used to describe operations of this kind and to signify hardware optimized for these applications, e.g. TensorFlow, Tensor-Processing-Unit, etc.

emissions [10], although this share will decline as grids decarbonize [11–14]. While estimates of IT hardware embodied emissions have significant uncertainty [15–17], previous analyses of enterprise computing equipment found embodied carbon accounts for only 2%–8% of lifecycle emissions [18]. This operational dominance is likely applicable for AI hardware given its high utilization and power density compared to conventional servers. Life-cycle assessment of tensor processing units estimated that operational emissions accounted for 90% of total emissions in the baseline case, dropping to 70% when accounting for zero-carbon power purchase agreements [19–21].

Second, facility siting, and by extension the carbon and water intensity of electricity supply [22] and the feasibility of free cooling, is the primary determinant of a data center’s overall environmental footprint [23]. As infrastructure energy use scales directly with IT load (i.e., cooling load scales with the waste heat generated by the IT hardware), accurate assessment of IT electricity use is fundamental to understanding both total facility energy consumption and environmental impact. Accordingly, this work prioritizes developing robust estimates of IT power demand during AI training.

2. Estimating AI electricity use

Historically, most assessments of AI related energy use have focused on the training of AI models. This in part stems from the distinctive profile of the physical hardware: training occurs on specialized, heat-dense processors in hyperscale facilities. While these scale and finite boundary characteristics are often perceived as making it straightforward to model and attribute energy use to the training of a particular model, in reality this process is considerably more complex and challenging than commonly understood [24]. It also reflects industry norms; hyperscale cloud providers typically bill ML training customers according to the number of accelerated chips they provide and their hours of use. Thus, GPU-hours serves as industry shorthand for model scale alongside parameter count.

One approach to modeling AI training energy use is based on empirically measured power draw of key hardware components [25–28] multiplied either by total mathematical operations [29] or by GPU-hours [30, 31]. As models have increased in size, so too have the upfront capital costs of training [32], and the ability to operate these facilities efficiently has become a key source of competitive advantage with publicly available information becoming scarce. Methodologically rigorous estimates of AI energy-use often rely on proprietary power draw data [11, 33, 34].

In practice, many estimates of AI training power consumption multiply the manufacturer rated maximum power draw (also referred to as *thermal design power* or (TDP)) of AI chips [35] or servers [36] by the reported training time (typically in GPU-hours) of a given model [37, 38]. This reflects data availability, as real time power measurements are not typically retrievable months or years after training has occurred [30], and multiplying GPU-hours by chip TDP is a straightforward modeling approach. These models will then multiply this output power by a parameterized PUE to arrive at a total energy footprint. These studies typically link their power estimates to a spatial-temporal analysis of carbon emissions [39]. Meta estimates the carbon footprint of training their LLAMA models through this chip TDP and GPU-hours approximation [40, 41]. Calibrating these estimates with empirical grounding serves to shed further light on the energy footprint of AI applications, and improve model accuracy [42]. While performance benchmarks have been proposed and implemented in the industry [43], training benchmarks have not kept pace with commercial system scale up [44].

Scholars typically either report the individual chip level energy performance [45, 46], or aggregate facility level power draw [35]. Data center energy use model validation requires chip, node, and system-level power measurement [26], and while GPU-level energy performance is well specified [33], nodes and clusters are not. Reliable documentation of node-level hardware performance for current generation workloads and servers is scarce, meaning even knowledgeable industry actors having to rely on rules-of-thumb in their estimates of in-use node power draw [47]. Our work addresses this gap for current generation hardware by developing an empirically calibrated statistical model of H100 node power draw during training. While this work primarily focuses on the energy footprint of training, see Appendix B for an overview of inference energy use and associated literature.

This paper makes the following primary contributions to the field: We analyze an integrated dataset of empirical measurements of AI training power demand on a single node combined with open-source training data on comparable workloads across single and multi-nodal training configurations. Through this analysis, we find that actual power draw consistently remains well below thermal design TDP, with even computationally intensive workloads drawing on average no more than 76% of TDP. We then use these data to develop an empirically calibrated statistical model of 8-GPU H100 DGX node-level power draw based on floating-point operations (FLOPs) per node, which achieves 11.5% mean absolute percentage error (MAPE) in predicting training workload power draw across our test

workloads—significantly outperforming TDP-based approaches which result in errors of 27–37%. Our analysis revealed distinct power signatures between convolutional neural network (CNN) and transformer architectures and characterized how node-level power demand varies with training configuration. Finally, we discuss these findings in the context of the energy system, identifying key drivers of uncertainty in future efforts to assess the energy footprint of AI computation.

3. Methods

Our empirically calibrated statistical analysis relies on a dataset constructed by combining two complementary sources of AI training power measurements. The first consists of detailed empirical measurements collected at Brookhaven National Laboratory (BNL), which include time-series data of node-level power demand on an 8 NVIDIA H-100 DGX during AI training benchmarks, as well as node idle power and GPU stress test data. Complete documentation of these measurements and the raw time-series data are available in Latif *et al* [48]. Those measurements were collected on a single-node, and so may not generalize to the multi-nodal training parallelization which defines the industry [49, 50], where data communication introduces additional complexity [51]. To address this limitation, we integrated the BNL data with open-source multi-node training data for comparable workloads, creating a dataset that spans both single-node and distributed training environments. Using these data, we developed a novel statistical model of node-level average power demand during AI training.

Training power draw data were collected from MLPerf® Training v4.0 [52] power measurement submission by Sustainable Metal Cloud (SMC)¹⁰. Initial data were retrieved on August 12th, 2024, from run IDs 4.0–0090 through 4.0–0096 (excluding 4.0–0093). We selected these open-source data due to the mixture of workloads and training configurations. We only analyzed the training runs which had a direct workload proxy to the BNL data, namely image classification (Resnet) and NLP (GPT3-175 and Llama). Training workloads without analogues in the BNL data we included in the validation set.

The hardware configuration of the SMC nodes does differ from the node at BNL. Both are dedicated DGX accelerated AI training servers with 8 NVIDIA H100-SXM5-80 GB GPUs. The manufacturer rated TDP for these systems is 10.2 kW [53]. SMC nodes employ two Intel Xeon Platinum 8462-Y model processors as supervisory CPUs, and are specified as containing 2000 GB of memory per server. In contrast, the BNL model contains 1500 gigabytes of memory and the supervisory CPUs are AMD EPYC 9354. Additionally, the SMC nodes are connected by NVIDIA specified Infiniband interconnect fabric. These are the differences available in publicly available documentation, without detailed bills-of-material we can only approximate differences between the hardware. Additionally, SMC nodes recorded power measurements approximately every two seconds, compared to the 5 min interval for the BNL data. While non-trivial, we can account for these differences through our choice of statistical techniques.

When error terms are correlated within clusters but independent across clusters, then conventional standard errors, which assume independence between all observations, will contain statistical bias [54]. We know our data contains some of these confounding factors, such as the differences in hardware configurations. Cluster-robust standard errors are designed to allow for correlation between observations within cluster [55]. Additionally, an unweighted regression of our data would in practice weigh the estimator towards the workloads with the fastest measurement interval and most nodes. To address these issues, we performed a weighted regression with robust clustered standard errors. Note that clustered standard errors do not affect the estimated parameter, just the estimation of the errors. We will now discuss our model specification and interpretation in greater detail.

4. Model specification

Our objective was to develop a statistical model that could predict node-level power-draw across different AI training workloads and system configurations. To develop this model, we began by considering the physical structure of power consumption in an AI training server. For a more detailed discussion of AI specialized IT hardware and facility characteristics see appendix A.

A server consists of several components that contribute to total power consumption. These components include logic elements that perform computations, memory systems that store data and intermediate results, and interconnect fabric that enables communication between them. When training an AI model,

¹⁰ MLPerf® Training v4.0 Power measurement benchmark. Retrieved from https://github.com/mlcommons/training_results_v4.0/tree/main/smc/results on 12 August 2024, entries 4.0–0090, 4.0–0091, 4.0–0092, 4.0–0094, 4.0–0095, 4.0–0096. The MLPerf name and logo are registered and unregistered trademarks of MLCommons Association in the United States and other countries. All rights reserved. Unauthorized use strictly prohibited. See www.mlcommons.org for more information.

the primary computation occurs in the parallel logic hardware, in our case the GPUs, which account for the largest share of power demand in a node [56]. Note that NVIDIA specified DGX servers contain 12 fan modules ‘behind the meter’. While not strictly IT load in the ideal theoretical definition PUE, for the purposes of metering the power supplied to a server or rack, this demand is bundled with the computational components.

At the most basic level, a server’s total power demand (Y) can be understood as the sum of power consumed by its constituent components:

$$Y = \sum_{i \in C} P_i. \quad (1)$$

Let set C represent the power-consuming components in a server: GPU accelerators (G), supervisory logic (U), memory and storage systems (D), interconnect (I), and cooling fans (F),

$$C = \{G, U, D, I, F\}. \quad (2)$$

While each component in set C could theoretically be measured independently, our data records aggregated power measured at the node level. The total power demand of computational hardware fundamentally scales as some function of the number of operations it performs (x) [57, 58]. For each component i in set C , we can express this relationship as:

$$P_i = \phi_i(x). \quad (3)$$

Logic chips, as the physical substrate performing the operations, have the most intuitive connection between operations and power draw, though the practical scaling relationship demonstrates complexity [59]. The other components in a server¹¹ can have more complex relationships with computational load. In some cases, such as embedded fans, these relationships are theoretically straightforward: cooling load scales with heat generation, which results from resistive losses in electrical components, creating a linear relationship between single-speed fan power and the electricity consumed by other components (assuming no external cooling infrastructure). This is a first order approximation, for a more detailed discussion of fan power, see appendix C.

The power demands of memory and interconnect components have more complex scaling behavior. These components both enable and respond to logical operations—memory must store inputs and intermediate results, while interconnect facilitates data transfer between components. This means they may scale independently with total operations, they may be dependent on logic chip computation, or they may be a mixture of direct or indirect dependencies. While future work could illuminate these complexities through component-level submetering, for our purposes modeling aggregate node power as a function of computational intensity captures these interactions while remaining tractable.

Therefore, the total power supplied to a server can be expressed as:

$$Y = \sum_{i \in C} \phi_i(x). \quad (4)$$

A finite piece of hardware can only perform some maximum quantity of operations within a given time [60]; more prosaically there would be some physical limit of computations where a processor would generate more heat than cooling systems could dissipate, causing thermal runaway and destroying the hardware. While individual components may scale differently with computational load—some linearly up to a hard limit, others asymptotically approaching a maximum—all must eventually reach some finite maximum power. Therefore, for any component i :

$$P_i \leq P_{i,\max}. \quad (5)$$

The exact functional form of $\phi_i(x)$ may vary by component type [59], see appendix C for a more detailed discussion, but the aggregate node power demand will be bounded by the sum of these component maxima at computational saturation, a quantity we define as (P_{comp}):

$$Y \leq \sum_{i \in C} P_{i,\max} \quad (6)$$

¹¹ In the case of an AI server this includes the supervisory logic devices which coordinate different server elements but do not themselves conduct the training computations.

$$P_{\text{comp}} = \lim_{x \rightarrow \infty} \sum_{i \in C} \phi_i(x) = \sum_{i \in C} P_{i,\text{max}}. \quad (7)$$

There are several candidate functional forms which could describe this system. The simplest specification assumes the server begins at some idle power (P_{idle}) at zero computational demand, and then approaches some asymptotic limit as (P_{max}) as computational demand increases. The difference between P_{max} and P_{idle} is then P_{comp} , with the parameter α governing the steepness of the transition:

$$Y = P_{\text{idle}} + P_{\text{comp}} \frac{x}{\alpha + x}. \quad (8)$$

We hypothesize that the actual power behavior follows a sigmoid function, as illustrated by equation (9). In this formulation, total power (Y) depends on the baseline power draw of an idle server (P_{idle}), the maximum additional power draw from active components (P_{comp}), the quantity of operations being performed (x), an operations threshold (x_0) governing the midpoint of demand saturation and a parameter (k) controlling the steepness of transition at which active power approaches the saturation value. Y , P_{idle} , and P_{comp} would be in units of Watts, x and x_0 would be in units of operations, and k would be dimensionless,

$$Y = P_{\text{idle}} + P_{\text{comp}} \frac{e^{(x-x_0)/k}}{1 + e^{(x-x_0)/k}}. \quad (9)$$

This specification captures several important physical characteristics of the system. The offset parameter x_0 represents a threshold of computational operations that can be performed using hardware resources already active for system maintenance and monitoring. Below this threshold, the marginal power demand for additional operations is minimal, as these calculations can be performed by otherwise underutilized cores which have been activated to perform for basic system functions. The parameter k determines how sharply power demand increases once operations exceed this threshold, reflecting how quickly the system transitions from idle to active states.

While the sigmoid model might better represent the underlying physics of the system, particularly at low computational loads, it requires estimating additional parameters that present challenges with our limited empirical dataset. Each additional parameter introduces greater uncertainty and increases overfitting risk, especially when working with measurements across heterogeneous hardware configurations. The multicollinearity between transition parameters (k and x_0) further complicates estimation, as different combinations can produce similar curve shapes, creating ridge issues in the optimization landscape.

Given these constraints, we prioritize robust estimation of the asymptotic power limit (P_{max}), which directly informs infrastructure planning and energy forecasting. Since both functional forms converge to similar asymptotic limits as computational demand saturates, their practical difference lies primarily in how they model the transition from idle to active states. To determine which specification better serves our purpose, we evaluate both the asymptotic and sigmoid models using out-of-sample prediction accuracy as our primary metric, focusing specifically on each model's ability to generalize across different AI architectures and training configurations. This approach emphasizes practical utility for energy forecasting rather than theoretical elegance.

We selected FLOPS as our measure of computational intensity. While parameter count provides a consistent measure of model scale, it inadequately captures the computational complexity that drives node-level power demand. Parameters fail to account for architectural differences in operations per parameter, hyperparameter impacts, parallelization strategies, activation functions, memory access patterns, and communication overhead. Another potential proxy for computational demand is the expected required *FLOPS* to train a given model (M). FLOPS directly represents the arithmetic operations required during training, providing a more precise measure of computational demand. For transformer architecture models, we adopted the established methodology from Narayanan *et al* (2021), which accounts for the specific computation patterns in these architectures, as shown in equation (10):

$$M_{\text{LLM}} = 96 \times B_{\text{global}} \times S \times L \times H^2 \times \left(1 + \frac{S}{6H} + \frac{V}{16LH}\right) \quad (10)$$

where B_{global} is the global batch size, S is sequence length, L is the number of layers, H is hidden dimension size, and V is vocabulary size. The coefficient 96 represents six operations per non-embedding parameter in both forward and backward passes ($2 \times 3 \times 16$), as empirically determined by Narayanan *et al* (2021). Note as is documented in the configuration files, global batch sizes for the language models were adjusted according to parallelization hyperparameters. Global batch size (B_{global}) was selected according to the relationship defined in equation (11):

$$B_{\text{global}} = B_{\text{mini}} \times \frac{\text{Total GPUs}}{\# \text{TP} \times \text{CP} \times \text{PP}} \quad (11)$$

where B_{mini} is the minibatch size, TP, CP, and PP are tensor, context, and PP respectively. These parallelization strategies distribute computation across hardware resources in fundamentally different ways: TP splits individual operations across devices, CP processes different sequences independently, and PP assigns different network layers to separate devices. Note that the benchmark submission LLM workloads adjusted global batch size based on their parallelization choices, creating an inherent interaction that cannot be isolated. While this collinearity prevents modeling each parallelization strategy's independent effect, our FLOPS-based computational proxy incorporates these factors through their collective impact on the derived global batch size calculation.

For CNNs, we calculated FLOPS based on the computational requirements of specific architectures, adjusted for image dimensions and accounting for both forward and backward passes [61], as defined in equation (13):

$$M_{\text{CNN}} = 3 \times M_{\text{image}} \times \left(\frac{I_{\text{train}}}{224} \right)^2 \times B_{\text{global}} \quad (12)$$

where M_{image} is the required FLOPS per image per forward pass for a particular model given a standard 224×224 input, I_{train} is the sidelength of the image in the training set, and B_{global} is the global batch size. Image size was sourced from the workload documentation, and M_{image} was sourced from the research literature [62]. We then divided these estimates of the required operations for an individual iteration of a given model by the number of nodes to get the effective FLOPS-per-node in a given workload (M_{eff}), which we use as our computational proxy (x). This metric enables meaningful comparison across architectures by capturing how model complexity, batch size, and parallelization jointly influence node-level power demand. All architectural parameters (hidden dimensions, layer counts, etc.) were sourced directly from the relevant model cards or benchmark workload documentation, while training hyperparameters (learning rate, batch size, etc.) were extracted from the benchmark configuration files. It is also noteworthy that effective FLOPS may not be ideal, as the energy intensity of operations can vary at a macro-system [42] and chip-level operational [63] level. A summary table of the architectural and hyperparameter values used for each workload is available in appendix D. A review of the literature on empirical scaling laws suggested that computational demand scales with the log of computational intensity [64–66]. This is the underlying physical model we statistically approximate: average node-level power draw ($\hat{Y}$) approaches an asymptotic limit, the sum of idle power and estimated active power (β_{comp}), as the log of computational intensity (x) increases. This asymptotic model specification is shown in equation (13):

$$\hat{Y} = P_{\text{idle}} + \beta_{\text{comp}} \frac{\log(x)}{\alpha + \log(x)}. \quad (13)$$

Note that this is an estimate of IT power, and does not include power switching losses, cooling load, or any other infrastructure energy use. While these are key considerations for assessing the environmental footprint of a given data center or AI model, they scale with IT demand [67]. We compared these IT power estimates to the sigmoid specification in equation (9).

Each of the nine workloads differed in the time required to complete training. The SMC data were sampled every 2 s while BNL measurements were taken at 5 min intervals, creating potential bias. Rather than discard data through downsampling to standardize measurement frequency, we preserved all observations and addressed these sampling biases through proportionate weighting scheme. We applied a weighting factor of $1/n$ to each observation within a workload, where n represents the total number of observations for that particular workload, ensuring that the cumulative weight of each workload summed to 1 regardless of its sampling frequency or duration. This ensures each training run contributes equally to our parameter estimates regardless of its duration, sampling frequency, or number of nodes. To evaluate the cluster robustness of our coefficients and standard errors, we clustered by individual workload.

We took the raw .csv and .txt time series of the power draw for each node training each of our workloads, and read them into statistical software, giving each individual node a unique identifier. Measurements were time-tamped according to time since training began. Interconnect power use in multi-nodal systems was allocated by dividing the total interconnect power use for each workload reported in the MLPerf submission by the number of nodes conducting training. For the transformer architecture models (GPT-3 and Llama), we sourced parallelism factors and batch size data from the MLPerf configuration files accompanying the SMC submissions, and from the experimental setup documentation for the BNL runs. Our final data set contains 11 6646 individual node-power draw observation across 9 different workloads. We performed the regressions in R, with the relevant code and our raw data available in [repository link]. To evaluate model predictive performance, we estimated the power draw of

Table 1. LOOCV coefficient stability and cluster-robust significance.

Cluster robustness results	Asymptotic model	Sigmoid model	
	α	k	x_0
Parameter estimate	5.55	1.12	11.46
Standard error	1.31	0.09	0.68
t-value	4.25	3.6	16.66
p-value	$<2 \times 10^{-16} ***$	$2.93 \times 10^{-7} ***$	$<2 \times 10^{-16} ***$
LOOCV summary	α	k	x_0
Mean	5.56	1.01	11.24
StdDev	0.43	0.4	0.72
CoV (%)	7.7	39.6	6.4

workloads and configurations not represented within our training set: a later submission to MLPerf for LLM training by a different submitter, and a UNET medical image classifier within the SMC submission.

4.1. Model selection and validation

Given the non-linear nature of our model specifications, a direct estimation approach presented convergence challenges. Attempting to simultaneously estimate all parameters across both the asymptotic and sigmoid functional forms resulted in inconsistent convergence, a common issue in non-linear regression with interdependent parameters. Our non-linear model specifications present parameter identification challenges due to strong interdependence between shape parameters (α , k , x_0) and magnitude parameters (P_{idle} , P_{comp}). When estimated simultaneously, these parameters often result in multiple local optima with similar fit quality but dramatically different parameter values. This ‘ridge’ in the optimization landscape leads to unstable convergence and physically implausible parameter estimates.

To address this challenge, we implemented a two-stage estimation procedure: first constraining magnitude parameters to physically measured values while estimating shape parameters, then using those stabilized shape parameters to refine our final model. This approach prioritizes parameters with direct physical interpretation while ensuring numerical stability. Our sequential fitting strategy is analogous to profile likelihood methods, where parameters of primary interest are estimated by constructing a profile of the likelihood function over nuisance parameters [68].

We incorporated some physical constraints into our initial model based on values measured in Latif *et al* (2025) for idle power and the GPU burn. The idle power for an air cool node with a rear-door heat exchanger at 24 °C was measured to be a stable value of 1.8 kw. During a combined stress test of the GPUs and CPUs on the node at their maximum utilization, the node stably drew 8.4 kW. Accordingly, in our models we constrained the maximum power at 8.4 kw, based on the empirical measurements from the GPU burn test. This represents an experimental proxy for the maximum possible utilization workload. With these constraints established, we proceeded to estimate the remaining parameters: the steepness coefficient (α) for the asymptotic model and the inflection point (x_0) and steepness (k) for the sigmoid model. This approach reduces dimensionality improving parameter identifiability.

To evaluate generalizability, we employed a three-part validation strategy:

- (1) Cluster robustness tests using heteroskedasticity-consistent standard errors grouped by workload type to account for within-workload correlation
- (2) Leave-one-out cross-validation (LOOCV) to assess steepness parameter stability across different architectural paradigms and training configurations
- (3) Out-of-sample validation of final model specifications using out-of-sample workloads including an independently submitted MLPerf LLM training run and a UNET medical image classifier

To validate our estimates, we performed a cluster robustness test, treating each individual workload as a distinct cluster. This accounts for potential correlation within similar workloads while allowing independence between different architectural paradigms and configurations. The coefficient estimates for our population-level steepness parameters were cluster-robust significant ($P < .01$) as shown in table 1, confirming that our parameter estimates remain robust when accounting for within-workload error correlation.

To further validate the robustness of this approach, we conducted LOOCV, systematically excluding each workload from our dataset and re-estimating the steepness parameters. This procedure assesses how sensitive our parameter estimates are to the inclusion or exclusion of specific workloads. The full

LOOCV summary table showing parameter stability as individual workloads are excluded is available in appendix D.

The LOOCV results demonstrate strong parameter stability across most workloads, with coefficients of variation (CoV) of 7.7% for the asymptotic model's steepness parameter and 6.4% for the sigmoid model's inflection point. However, we identified BNL—Resnet 1 as a potential outlier, with parameter estimates diverging notably from the rest of the dataset, indicating a problematically high-influence cluster within our sample.

This workload exhibited low GPU utilization: 36% on average as compared to 77%–93% in the other BNL runs [48]. This is likely due to the small batch size causing frequent memory transfers that throttle GPU performance [69, 70]. This creates a fundamentally different power-computation relationship than our model aims to characterize, as the hardware is not compute-bound but memory-bound. Including this outlier would bias our model toward a power-computation relationship not representative of the performance-optimized configurations typical in production environments.

Given the high coefficient stability across the other workloads, we excluded the BNL—Resnet 1 workload from our final parameter estimation to avoid biasing our model toward this atypical operating condition with a known experimental confound. Additionally, the SMC—Llama-70 B [8] workload shares identical architectural, hyperparametric, and nodal characteristics with one of our validation workloads. To avoid data leakage, we excluded this workload from our training set, including it instead in the validation group. After these exclusions, we recalculated the population values for steepness parameters to use in subsequent estimation phases: $\alpha = 5.11$ for the asymptotic model and $x_0 = 9.91$. Given the high CoV of k , we elected to fit our sigmoid models leaving k unconstrained, and they were able to successfully converge to physically reasonable values.

With stable steepness parameters identified through cross-validation, we proceeded to estimate three final model specifications. We fit dataset-wide asymptotic and sigmoid models using the established steepness parameters. Additionally, to account for architecture-specific power behaviors observed in preliminary analysis, we developed an extended asymptotic specification incorporating fixed effects for model architecture (transformer vs CNN). This architecture-specific model maintains the same functional form but allows separate magnitude parameters for each architecture type, potentially capturing the distinct power profiles which vary by the underlying model architecture [42].

5. Results

5.1. Workload characterization

Resnet [71] models are CNN, while Llama2 [40] and GPT-3 [29] are each Transformer architecture DNNs. All workloads sourced from Latif *et al* are preceded by the identifier BNL, and all other workloads were sourced from SMC. The BNL workloads include two configurations of Resnet, which differ by batch size. BNL-Resnet 1 refers to the 512-image global batch workload, while BNL-Resnet 2 corresponds to the 4096 batch size. The SMC training runs concluded upon reaching MLPerf-specified quality targets. For image classification, ResNet training continued until achieving 75.90% accuracy on ImageNet. The GPT3 language model trained until reaching a log perplexity of 2.69, while Llama-70 B is fine-tuned with the LoRA method on the SCROLLS GovReport dataset concluded at a cross entropy loss of 0.925. A full inventory of summary statistics for each workload is shown in table 2. Boxplots of all sampled power measurements for each workload are displayed in figure 1.

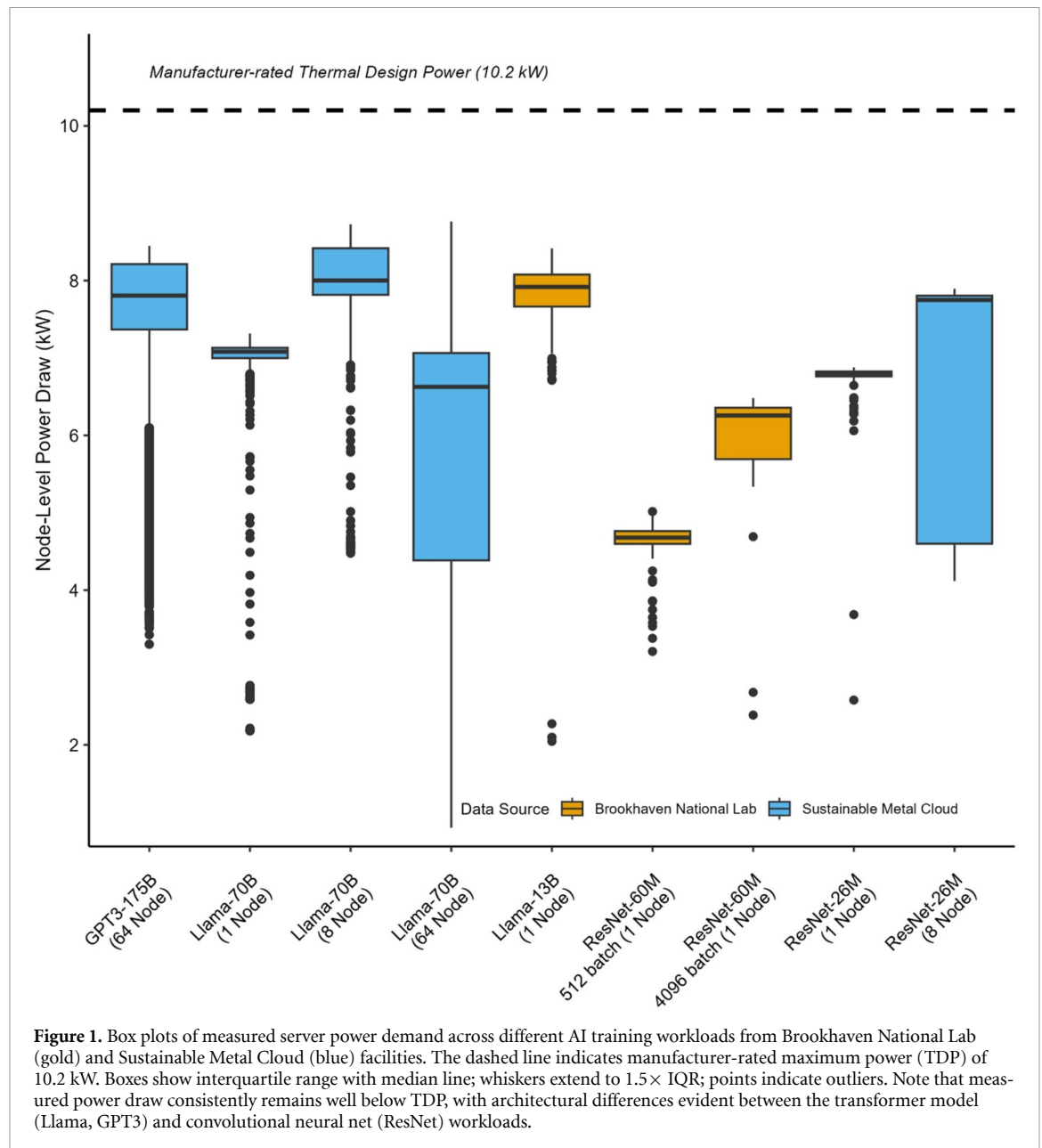
While multi-nodal training produced the highest measured per-node maximum power, the node-level average for the most computationally dense workloads was consistent across single and multi-nodal training. In fact, the highest average node power (7.79 kW) was observed in the single node BNL training of Llama-13 B. Note that the average and maximum power draw measured in each workload never approached the manufacturer rated maximum of 10.2 kW.

Models of different architectures differ in their load profile. Shallow CNNs display a flat power demand time-series, with a single, uninterrupted period of high computational intensity. This is consistent with the load profile of physical simulation and other conventional HPC workloads [72]. An illustrative example of this load shape is the image classifier training shown in figure 2. In contrast, DNN models display characteristic square-wave load, with periods of higher demand punctuated by troughs at both the chip and the node level, as shown in figure 3.

The characteristic DNN waveform reflects the parallelization of the computation. During training, each accelerator only operates on a partition of the model using a subset of the training data. While individual GPUs can achieve high utilization during active computation, they briefly enter lower-power states while waiting for results from other parts of the system. These troughs in power demand occur at multiple scales. At the chip and node level, GPUs may be throttled while waiting for results from slower

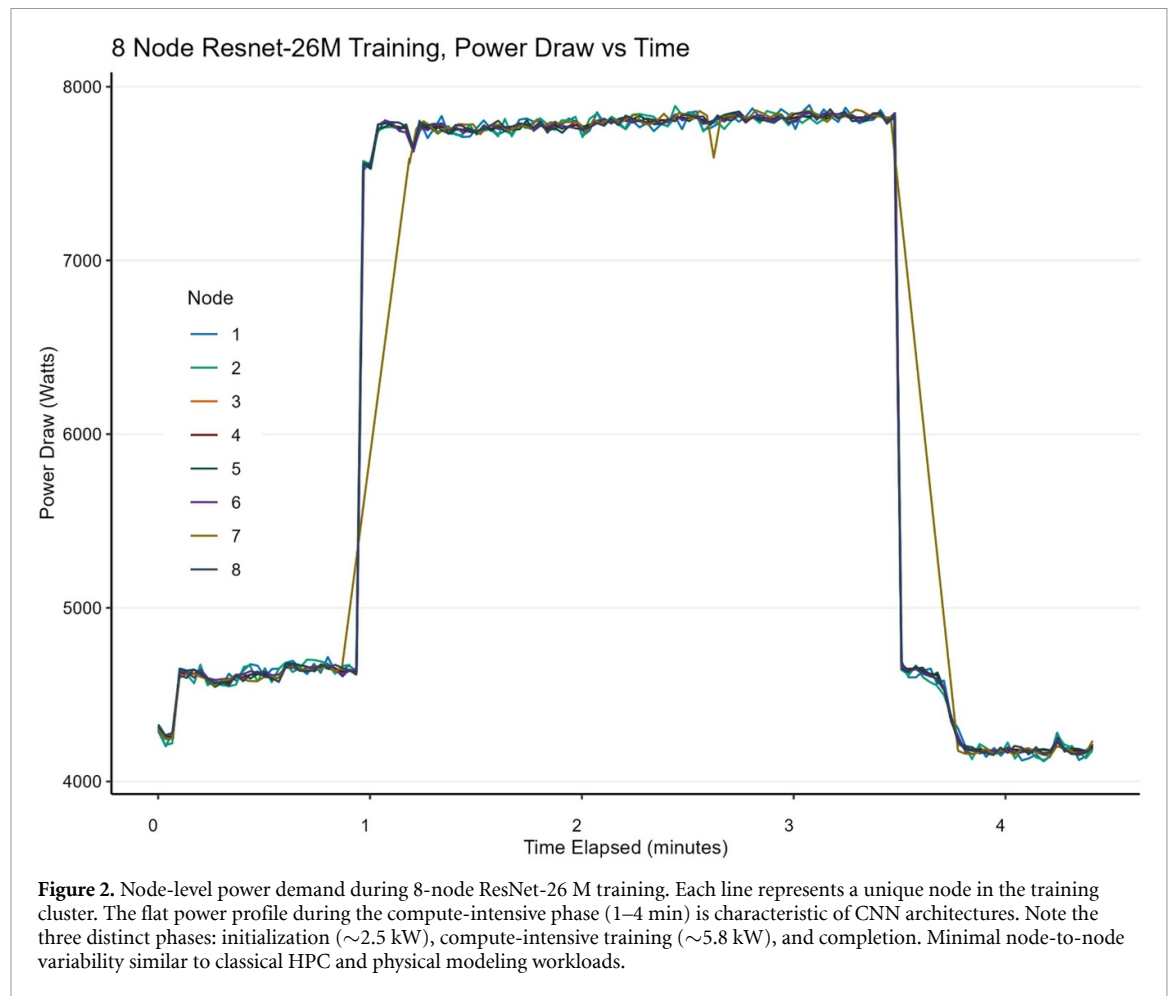
Table 2. Summary statistics for each individual workload in our dataset. All systems use NVIDIA H100-SXM5-80GB GPUs (8 per node). B = billion parameters, M = million. Batch refers to the global batch size. For each workload, average power (P_{avg}), maximum power (P_{max}), and standard deviation (SD) are reported in kilowatts (kW). Power measurements are per-node, energy values represent total IT system consumption including estimated interconnect power. Global batch size (Batch) values are sourced from the relevant configuration files. FLOPS are the estimated floating point operations per iteration, calculated according to equation (10)–(12), and FLOPS/node is our effective FLOPS-per-node computational demand estimate. BNL indicates Brookhaven National Laboratory data, while SMC indicates open-source benchmarking data from Sustainable Metal Cloud. Columns proceed left-to-right from workload configuration and identifiers to measured values.

workload	arch.	param.	batch(global)	nodes	flops	flops/node	p_{avg} (kw)	P_{max} (Kw)	sd(kw)	duration(hours)	it energy(kwh)
smc-gpt3-175 b[64]	DNN	175B	2048	64	6.01×10^{18}	9.40×10^{16}	7.67	8.45	0.73	0.95	464.94
smc-llama-70 b[64]	DNN	70B	64	64	1.47×10^{17}	2.29×10^{15}	5.91	8.76	1.52	0.03	12.10
smc-llama-70 b [8]	DNN	70B	8	8	1.83×10^{16}	2.29×10^{15}	7.73	8.73	1.11	0.09	5.43
smc-llama-70 b [1]	DNN	70B	8	1	1.83×10^{16}	1.83×10^{16}	6.78	7.32	1.05	0.5	3.39
bnl-llama-13 b [1]	DNN	13B	8	1	6.03×10^{18}	6.03×10^{18}	7.79	8.42	0.61	8	62.36
smc-resnet 1 [8]	CNN	26 M	3200	8	3.46×10^{13}	4.32×10^{12}	6.36	7.89	1.66	0.07	3.75
smc-resnet 2 [1]	CNN	26 M	3200	1	3.46×10^{13}	3.46×10^{13}	6.76	6.88	0.29	0.22	1.51
bnl-resnet 1 [1]	CNN	60 M	512	1	3.54×10^{11}	3.54×10^{11}	4.6	5.02	0.34	26.8	123.41
bnl—resnet 2 [1]	CNN	60 M	4096	1	2.83×10^{12}	2.83×10^{12}	5.76	6.48	0.11	5.25	30.15



computations elsewhere in the network. System-wide drops in power draw occur during key synchronization points: when partial results are checked for consistency across nodes, when parameter updates are propagated through the system, when updated weights are written to memory, and when new training data is loaded from storage into GPU memory. Future empirical research should collect detailed component-level sub-metered demand data during training runs to identify the composition of demand over time.

These results demonstrate the importance of empirical validation of training power demand. At the node level, the interactions between accelerator computational demand, memory, supervisory logic, and interconnect mean IT power demand is not steady state within deep learning workloads. Further, previous work has found that the discrepancy between empirically measured power demand on a wattmeter and software approximation differed according to accelerator [73, 74]. The optimization of node-power-performance vs throughput vs system energy performance is non-trivial [75]), see appendix A for more detail. These relationships evolve as the hardware does, as clusters grow and individual racks approach 100 kW of rated IT load [53] detailed empirical understanding of these interactions should keep pace. Component level data could enable system optimization, and detailed time resolution of power draw data could enable better assessment of the grid impacts of these facilities.



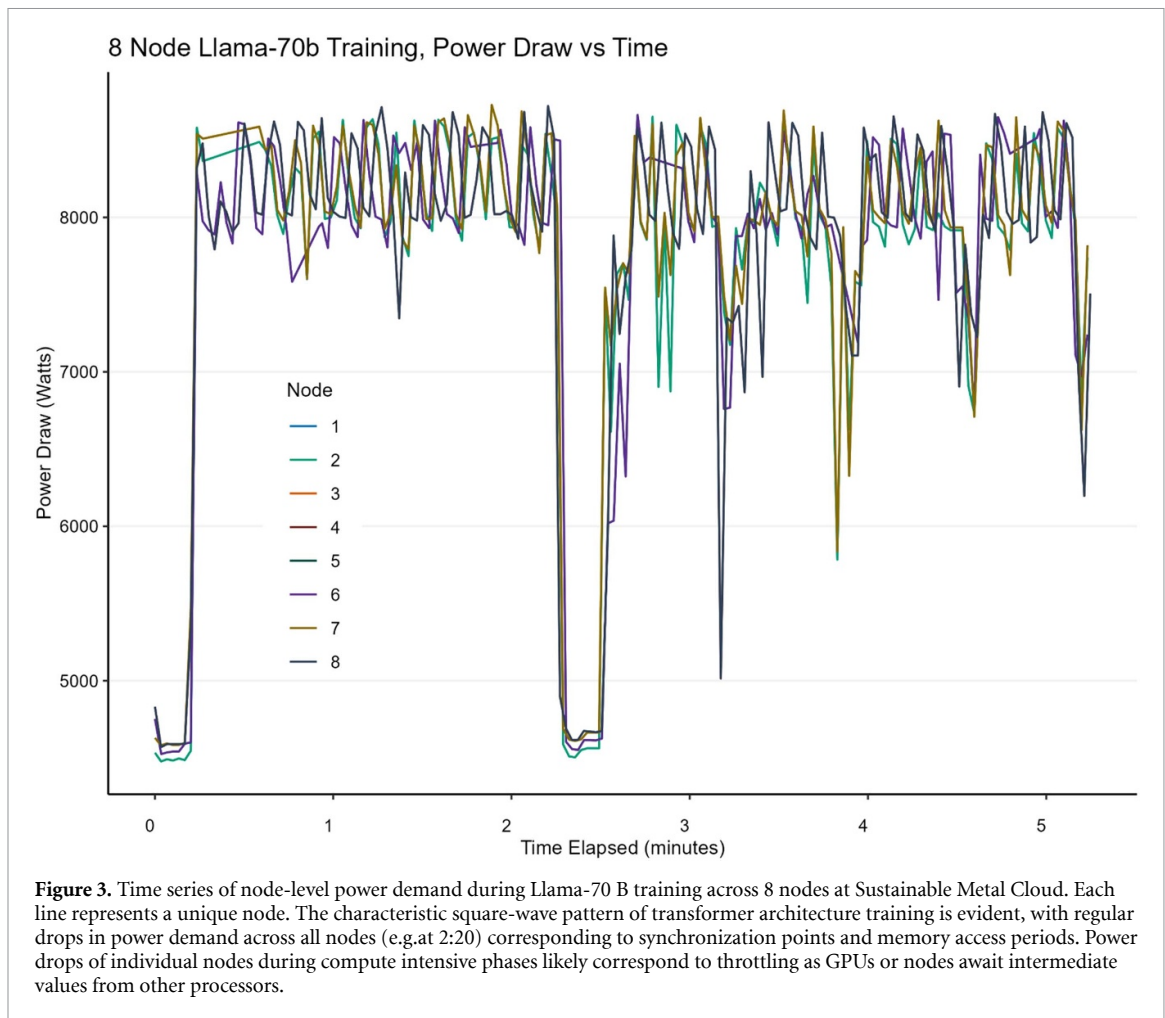
6. Statistical estimators and model performance

Our analysis evaluated three model specifications: a basic asymptotic model, an architecture-specific model differentiating between transformer and CNN architectures, and a sigmoid model. All models were fitted using the two-stage estimation procedure described in the methods, with fixed steepness parameters derived from our preliminary analysis: $\alpha = 5.11$, $x_0 = 9.91$, and a P_{idle} value of 1.86 kW. Table 3 presents coefficient estimates and statistical significance for each model.

To evaluate predictive accuracy, we assessed our models using both in-sample workloads and four out-of-sample validation workloads representing diverse configurations: two 8-node Llama-70 B implementations (from Dell and SMC) and UNet-3D medical image classifier workloads trained on either 1 or 9 nodes. This validation dataset spans both transformer and CNN architectures, includes single and multi-node configurations, and incorporates data from multiple sources, providing a robust test of model generalizability. For each workload, we compared estimated energy consumption (based on modeled power and average node duration) to empirically measured energy use to calculate MAPE. A full table of summary statistics for the validation workloads is available in appendix D.

While the sigmoid model achieved the best in-sample fit (MAPE = 9.34%), the architecture-specific asymptotic model demonstrated superior generalization to unseen workloads (MAPE = 5.39%). This suggests that accounting for architectural differences in power consumption patterns can improve workload power prediction. The superior out-of-sample performance of both the sigmoid and architecture-specific models compared to in-sample accuracy likely reflects the validation datasets' closer alignment to compute-optimal configurations. Overall, coefficients were stable across different specifications, with estimates differing by less than cluster robust standard errors. Note the comparatively large standard error in sigmoid active power estimate, and the lack of cluster-robust significance in the steepness estimate.

To assess model robustness, we conducted LOOCV, systematically excluding each workload and re-estimating model parameters. Table 4 summarizes the stability metrics across all models.



All asymptotic model variants demonstrated excellent parameter stability with CoV below 3.2%, indicating robust estimation regardless of which workload was excluded. The architecture-specific parameters maintained tight ranges despite architectural diversity in our dataset, with LLM compute power ($\beta_{\text{Comp_LLM}}$) consistently higher than CNN power ($\beta_{\text{Comp_CNN}}$) across all LOOCV iterations. The sigmoid model showed acceptable stability for P_{max} (CoV = 8.59%) but poor stability for the transition steepness parameter k (CoV = 42.14%), confirming parameter identification challenges in this specification.

Based on our analysis of cluster-robustness, predictive validity, and parameter stability, we selected the architecture-specific asymptotic model as our preferred specification. This model balances excellent parameter stability (CoV < 3.2%) with superior out-of-sample prediction accuracy (5.39% MAPE), while capturing power consumption differences between transformer and CNN architectures. The basic asymptotic model offers slightly better parameter stability but sacrifices prediction accuracy and fails to capture architectural differences in power behavior. The sigmoid model, despite strong predictive performance, suffers from parameter instability and identification issues, and a lack of cluster-robust significance in the steepness parameter, that limit its practical utility. The sigmoid model may better capture power behavior at very low compute intensities where theory suggests it better represents gradual resource activation, while both specifications perform similarly at high utilization. For intermediate utilization regimes, and when comparing workloads with different architectures, the architecture-specific asymptotic model provides superior predictive accuracy.

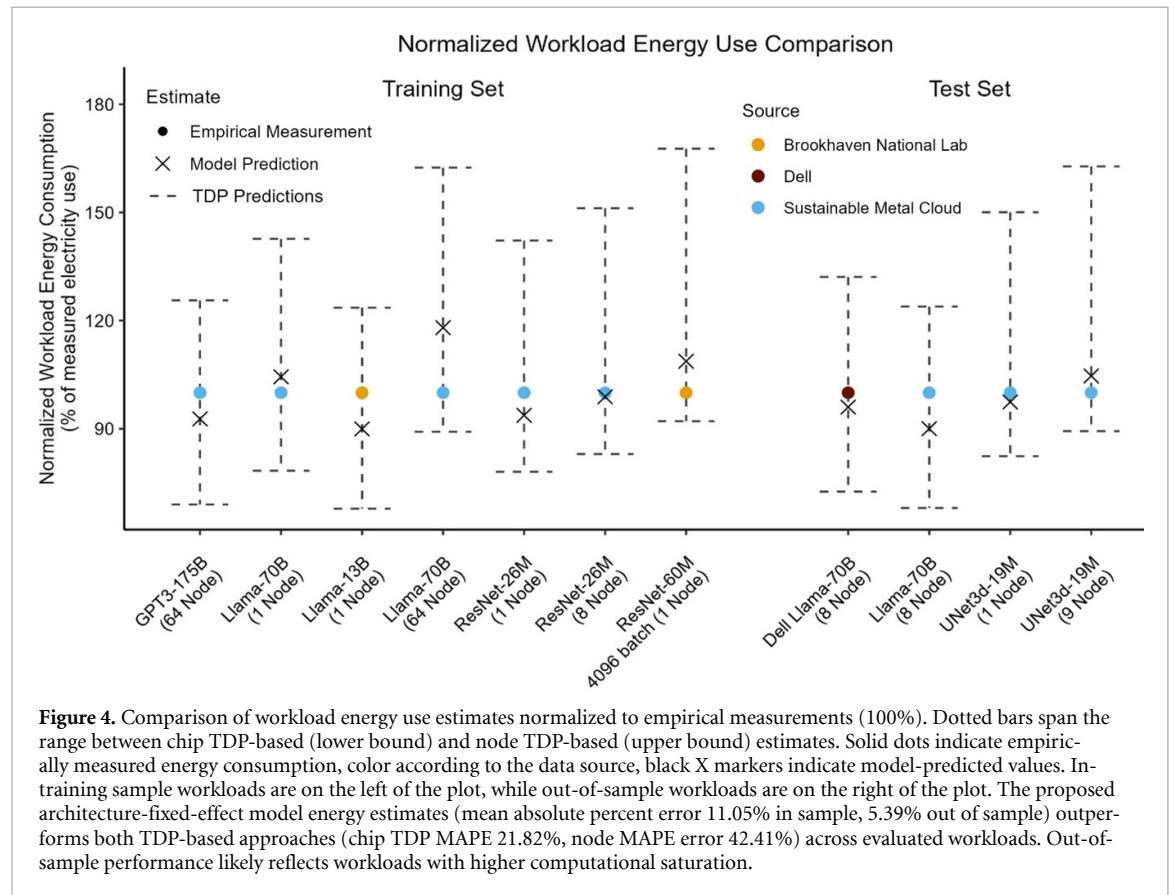
In terms of the performance of the statistical estimator, our goal was to better predict the total energy use of these workloads using this estimate of average node power demand while training. To evaluate this, we compare our empirically measured energy use for each workload to an estimate based on training duration and an approximation of node power. These approximations include our architecture fixed-effect model prediction, the TDP of the GPU as a lower bound (GPU-Hours X GPU TDP), and the TDP of the Node as an upper bound. The percentage errors of these three estimates of node power draw across our in-sample data are as follows: $\delta_{\text{node}} = 36.83\%$, $\delta_{\text{chip}} = 27.26\%$, $\delta_{\text{model}} = 11.46\%$.

Table 3. Robust weighted regression estimates of active power coefficients estimates, statistical significance, and prediction accuracy for three model specifications. The Asymptotic model uses a single power saturation parameter, while the Architecture-Specific model employs separate parameters for LLM and CNN workloads. The Sigmoid model implements a different functional form with transition steepness parameter k. All models were validated against four out-of-sample workloads. All significance and standard errors are the cluster-robust values. Stars denote significance codes, MAPE values show in-sample error followed by out-of-sample error in parentheses and demarcated by a dagger. Note the parameter k is dimensionless and is displayed with the others for convenience.

Model	COEF.	Estimate (kW)	STD. Error (kW)	T-Value	P-Value	Energy Mape (%)
Asymptotic $\hat{Y} = P_{idle} + \beta_{comp} \frac{\log(x)}{\alpha + \log(x)}$	β_{comp}	6.65	(.35)	19.04	< 0.001***	11.8(7.6 [†])
Architecture-fixed effect	$\beta_{comp \text{ LLM}}$	6.89	(.50)	13.65	< 0.001***	11.1(5.4 [†])
where β_{arch} depends on model architecture	$\beta_{comp \text{ CNN}}$	6.28	(.32)	19.46	< 0.001***	
Sigmoid $Y = P_{idle} + \beta_{comp} \frac{e^{(x-x_0)/k}}{1+e^{(x-x_0)/k}}$	β_{comp} k	6.94 0.19	(1.33) (.16)	5.23 1.19	< 0.001*** .235	9.3(5.5 [†])

Table 4. Parameter stability analysis across model specifications, showing means, coefficients of variation (CoV), and ranges when systematically excluding each workload. Lower CoV values indicate greater parameter stability.

Model	Parameter	mean (kw)	cov (%)	range (kw)
Asymptotic	B_{Comp}	6.6	2.0	6.4–6.8
architecture-specific	$\beta_{\text{Comp LLM}}$	6.9	3.2	6.6–7.3
Sigmoid	$B_{\text{Comp CNN}}$	6.3	2.5	6.0–6.6
	B_{Comp}	6.7	8.6	5.4–7.6
	k	0.23	42.1	0.14–0.46



In terms of practical utility, estimating workload energy use provides the most value. We evaluated each model's predictive performance using MAPE for both in-sample and out-of-sample predictions. Plots of the architecture-fixed-effect estimates of total energy use of both in-sample and out-of-sample workloads, normalized to the empirical measurements for each workload are shown below in figure 4.

7. Limitations

A main limitation to this research is the availability of empirical data for emerging hardware configurations. Next-generation GPU architectures (e.g. Blackwell B200) and newer rack-scale configurations such as NVIDIA's 72-GPU pods may exhibit different thermal and power distribution characteristics compared to the 8-GPU nodes we studied. With respect to cluster scale, our dataset spans 1–64 nodes and we observe node-level power maxima consistent across node quantities. Commercial AI training clusters comprising tens of thousands of GPUS, unless they are in emerging configurations like rack scale pods, are composed of nodes architecturally similar to those we measured. If large-cluster networking overhead introduces non-linear GPU throttling, then we would predict clusters capable of training larger models with fleets of more GPUs, but at lower chip and node level utilization than we observe in our data. Facility data is necessary to confirm generalization of these results or to characterize distinct behaviors at larger cluster scale.

Commercial operators face strong economic incentives to maximize hardware utilization, suggesting that production workloads should similarly saturate available computational resources. Unless network communication overhead scales in ways not captured by our multi-node measurements, our observed per-node power behavior should generalize to larger deployments. The 8-GPU server configuration remains the dominant unit of deployment for the time being and thus represents our most appropriate unit of analysis until empirical data from alternative configurations become available. Extension of these results through larger datasets incorporating more diverse workloads, emerging hardware configurations, and varied operational conditions represents an important avenue for future research.

Prior work has identified systematic discrepancies between GPU-embedded sensor readings and externally validated power measurements using calibrated instrumentation [73, 76]. While facility-level power monitoring through DCIM software and PDU metering typically provides more reliable data than onboard sensors [67], the temporal resolution and component-level granularity of these measurements could be improved through more sophisticated metering.

The interaction between IT load and cooling infrastructure represents another limitation. We prioritized estimating IT load as in energy-systems modeling, infrastructure and facility energy-use are typically extrapolated from this parameter. However, IT load and infrastructure are not fully independent, especially in compute-dense applications [77]. H100 GPUs operate with dynamic-voltage-frequency-scaling (DVFS), where voltage and clockspeed are adjusted to optimize energy or throughput based on workload characteristics and chip temperature [78]. This creates a bidirectional relationship between cooling efficiency and computational performance: more effective cooling enables higher operating frequencies and more energy-efficient transistor operation [79]. DVFS techniques can dramatically improve energy performance [80], with inference representing a particularly promising application [81, 82].

Direct-to-chip liquid cooling [83–86] or immersion cooling [87] could enable different power-performance tradeoffs by maintaining lower junction temperatures. This would allow GPUs to sustain higher clock frequencies without thermal throttling, potentially resulting in higher instantaneous power draw but shorter training times and potentially improved overall energy efficiency. Future work should systematically evaluate the impact of cooling technology on node-level power profiles across identical workloads, ideally using controlled experiments with the same computational hardware under different cooling regimes. Including direct measurements of cooling system energy consumption would support more accurate total facility energy projections and inform cooling technology selection for new AI training facilities, and would enable better characterization of the cooling technology impact on IT performance [88].

Additionally, the operational intensity of a given model (required FLOPS per parameter) is known to vary according to architectural choices. More sophisticated model specifications based on detailed relationships between hyperparameter choice, architectural characteristics, and workload configuration may enable more accurate prediction, conditional on their ability to generalize across workloads. The lack of component-level submetered power data also limits our ability to precisely attribute power draw to specific elements of the system during different phases of training. These limitations should be considered when assessing the implications of this work.

8. Discussion

To illustrate the importance of accurate power measurement, consider a hypothetical training cluster in Iowa containing 20 000 H100 GPUs across 2500 nodes—a scale consistent with current commercial clusters [89]. Consider all 2500 nodes to be training foundation model for 90 d, an approximate training duration of current commercial models. Assume the cluster has a PUE of 1.1, the value Alphabet reported for their facilities in 2023 [90], and ten percent facility power conversion and switching losses.

Our architecture fixed-effect asymptotic model predicts each node draws 7.3 kW. Over the 90 d training period, we estimate this cluster would consume approximately 49.0 GWh of electricity. Using Iowa's regional grid carbon intensity of 428 kg CO₂e per MWh [91], this training run would then emit 20 970 tons of CO₂e. However, if we had instead relied on the manufacturer-rated TDP of 10.2 kW per node, as many facility planners do, our estimate of total energy consumed would be 68 GWh. This 19 GWh difference in estimates represents approximately 8132 tons of CO₂e—equivalent to the annual emissions of 1900 passenger vehicles.

The carbon intensity value itself introduces considerable uncertainty. Using the EPA's non-baseload value of 806 kg CO₂e per MWh rather than the average produces estimates differing by 15 300 tons CO₂e. This demonstrates why accurate power measurement proves critical for both infrastructure planning and environmental impact assessment, as measurement uncertainty compounds across impact estimates. While this hypothetical facility's absolute energy and carbon footprint remains significant

regardless of the calculation method, effective planning and optimization demands precise operational understanding.

Uncertainty in the IT load for AI training facilities is not merely a theoretical concern; it creates tangible challenges for infrastructure providers responsible for electrical grid stability and capacity planning. The lack of transparency around actual power demand in the data center sector has historically resulted in numerous inaccurate claims and projections of energy consumption [92]. Empirical data to understand IT power demand under actual operating conditions was highlighted in a recent report to U.S. Congress [93].

In our conversations with electrical utilities and energy infrastructure planners, uncertainty in IT demand was a consistent pain point. As one grid infrastructure planner candidly expressed: ‘The most significant headache in working with [AI training facilities] is that I do not know how much [electricity] their IT will actually consume. When a facility claims it is 100 MW [rated IT load], how much demand is that really?’¹² This ambiguity compounds through multiple sources of uncertainty: the gap between rated and actual hardware power consumption, facility-specific cooling configurations (which can vary from traditional air cooling to more efficient direct-to-chip liquid solutions), and phased deployment schedules that may spread capital expenditures across years. Infrastructure planners must determine whether a facility marketed as ‘100 MW of IT load’ represents fully deployed capacity or includes planned expansions financed by initial operations. These uncertainties directly impact critical planning decisions: transformer and substation capacity requirements, transmission infrastructure sizing, and even generation resource adequacy assessments.

The rapid expansion of AI infrastructure need not jeopardize energy system reliability or the clean energy transition, but it requires both greater transparency in facility planning and careful management of actual operational demand profiles. The sheer scale of the workload means that operators are well incentivized towards efficiency [94]. Additionally, the up-front capital requirement and small supply of AI training hardware have resulted in a concentration of capabilities in the largest scale data center operators [92], service providers with demonstrated industry leading efficiency performance [94–96], though the emergence of AI native small-cloud providers may complicate this characterization.

The heat density of specialized AI training chips makes them ideal matches for the highest efficiency, liquid cooling systems [97], both direct-to-chip and immersion. Recognizing this reality, in 2023 Meta fundamentally redesigned their AI data center for liquid cooling infrastructure [98]. The duration of large-scale training runs means that AI training has low latency sensitivity [31]. This is not to say that individual elements of the AI training hardware stack are latency tolerant between one another, rather that the duration of training runs makes them insensitive to location derived network delays. This robustness combined with the post-training processing models go through means that these facilities have high degrees of freedom in site selection. As the bulk of data center environmental impact is site dependent [23], AI training centers are free to locate where the power is green and inexpensive, can be carbon-aware scheduled [31], and are strong candidates for demand response [99].

The relationship between cooling technology and energy efficiency requires some detailed consideration. While liquid cooling enables higher GPU operating frequencies and voltages, this can result in non-linear increases in power consumption due to the voltage-frequency characteristics of semiconductor devices. At the node level, liquid cooling may reduce energy consumption by replacing inefficient air-cooling fans, but the net effect on system energy use is complex. Users may leverage the increased computational throughput enabled by liquid cooling to run more intensive workloads, potentially creating a rebound effect in energy consumption. Alternatively, while liquid cooling might enable operating states with worse instantaneous energy efficiency (in terms of operations per watt), the associated performance improvements could reduce total training time and improve lifecycle energy use. Understanding these tradeoffs requires detailed empirical measurement, at the chip, node, cluster, and facility level.

Previous work projecting large growth in data center energy use did not adequately account for the improvements in data center energy efficiency, which enabled global internet traffic to grow by 550% from 2018, while data center energy use only went up 6% [94]. However, AI compute is fundamentally different from traditional HPC, and counting on data center efficiency to blunt the growth of AI energy use is not a robust strategy. The stakes for accurate forecasting are high—future data center electricity projections inform stakeholders on needed development for electricity generation and transmission to allow for strategic growth that benefits the environment and broader economy [100]. Overestimating future electricity demand may lead to development of electricity infrastructure unable to recoup costs, while underestimates can impede AI technology progress.

¹² Personal communication, Virginia electrical utility employee, April 2025.

The rapid power fluctuations observed during training workloads at the node level, results consistent with findings at the chip level [33], could stress the grid, requiring fast-ramping and grid-forming resources to ensure service stability [101]. These workload timeseries suggest that an 80 000 chip cluster would have in-training demand swings of approximately 24 MW in the space of 20 ms. Swings of such scale and speed are known to cause issues for local electric grids at HPC centers [102]. How problematic these transients will be to electrical infrastructure depends on their frequency and ramp rate, precise characterization of which requires sampling at greater resolution than our dataset or integrated GPU sensors [103]. These data would enable data center users to optimize and smooth power demands and grid operators to manage anticipated loads [104].

While reporting total energy use to train a given model is vital for assessing efficiency and hardware utilization, it is insufficient to characterize the strains these workloads place on electrical infrastructure. The temporal power demand characteristics, particularly high frequency demand fluctuation observed in transformer architecture training, create significant operational challenges [105]. As models grow in parameter count and training duration, and as training potentially becomes distributed across facilities, understanding these load profiles becomes critical for infrastructure planning and operation. High-resolution time series power measurements would enable several key optimizations: identification of synchronization bottlenecks, evaluation of different parallelization strategies, and opportunities for power-aware scheduling [106]. Additionally, detailed power data could inform the co-design of algorithms and hardware to better manage these distinctive load patterns, potentially enabling more efficient facility operation without sacrificing training or model performance. Long-run management will require a combination of GPU-level hardware interventions, software optimization, responsive facility energy storage [103], and grid side demand and frequency response efforts [104]. Even when large-scale outages are avoided, these rapid power fluctuations degrade local power quality through voltage sags and harmonic distortion, disrupting sensitive equipment at nearby hospitals, co-located data centers, and residential areas that share distribution infrastructure with training cluster.

9. Sources of uncertainty in AI energy use modeling

These results suggest several lines of research on AI related energy use. One is continued empirical effort to comprehensively characterize the energy performance of hardware running AI workloads. Benchmarking the energy performance of training across different cooling technologies is one such approach. Alternatively, detailed component-level submetering would provide more granular understanding of subsystem utilization and could potentially enable workload optimization. As hardware and facilities evolve, empirical work must keep pace. Future analyses or benchmarks of training energy should include node or cluster level power demand time series and GPU utilization [92]. While more technically challenging [107], component sub-metering would provide greater still greater insight.

Future hardware generations may exhibit fundamentally different component power distributions within the server bundle. The transition from air-cooled to liquid-cooled GPU configurations, for instance, would dramatically reduce or eliminate the behind-the-meter fan power we measure (see appendix C), shifting the shape of the power-performance curve in ways our current model cannot predict. Another critical source of uncertainty in projecting facility-level energy consumption is the capacity factor for training hardware (the fraction of time training hardware operates at high utilization). While the high capital costs of AI accelerators create strong economic incentives for continuous operation, actual scheduling involves maintenance windows, workload transitions, debugging periods, and varying job sizes that prevent sustained peak utilization. Production job composition for a modeled cluster is likely mixed in terms of computational intensity, and there is some friction in scheduling, implementation, and dispatch. Each would reduce the energy footprint of a given cluster compared to a theoretical maximum.

ML research should expand to more systematically characterize the energy implications of architectural choices. The energy performance impacts of hyperparameter choice, multi-facility training, and parallelization across the 10 000 GPU clusters remain unclear. Understanding these relationships could inform model development—complementing the traditional metrics of accuracy and training time with energy performance. Energy systems research should develop more granular understanding of these facilities' load profiles and grid impacts, grounded in empirical power draw data. More detailed characterization of load variability and ramp rates would enable both better infrastructure planning and potential optimization opportunities. Every time a benchmark is run, energy and power should also be measured.

The nature and scale of demand for inference is a major source of uncertainty in long-term models of AI energy use, with a more detailed discussion available in appendix B. There are significant economic factors which underly scholarly projections of future inference energy use. One is the

rate of consumer adoption of AI related products beginning in 2023; by some measures ChatGPT had the fastest rate of consumer adoption in software history [108]. Additionally, the training cost structure encourages expansive deployment of AI models. Recall that training is performed in discrete generational batches, typically on a hardware stack dominated by upfront capital costs [109]. This training cost structure means that firms have a strong incentive to deploy trained models as widely as possible, to better amortize the upfront training cost. Inference costs will be passed through to the user, either directly through subscription fees or indirectly through monetization of inference related web traffic. Thus, while firms will seek to recoup model training costs through broad deployment and long-term use, inference queries will be driven by end user value perception. These structural factors give us directional insight into inference computational share, but scale remains an open question, as it will be driven by user demand.

Models of future inference energy consumption face interacting uncertainties: hardware performance, hardware composition, algorithmic efficiency, demand profile, and facility characteristics. The effective energy performance of the inferencing hardware stack is in part determined by composition, what chips are doing the inferencing. The representative facility and cooling system for inference will depend on this hardware stack, as well as the latency sensitivity of the workload. Demand for inference will inform both the total scale of the workload, and the latency sensitivity of this demand. Accurately predicting any of these characteristics in isolation is daunting, assessing all four as they mutually interact presents a profound challenge. Future work on modeling the energy footprint of inference should explicitly state the assumptions of these characteristics for each scenario, to ensure a clear understanding by researchers of the respective sensitivity.

10. Conclusion

Our empirical analysis demonstrates that AI training hardware operates well below manufacturer-rated power limits, with even computationally intensive workloads drawing no more than 76% of rated TDP. Through development of a novel statistical model relating computational intensity to node-level power demand, we achieved significantly improved accuracy in predicting training energy consumption compared to conventional TDP-based approaches. This more precise understanding of actual power dynamics enables better infrastructure planning and optimization while highlighting important differences between CNN and transformer architecture power signatures that must be considered in facility design.

The rapid evolution of AI capabilities and corresponding growth in computational demands makes accurate power measurement and modeling—and managing uncertainty—increasingly critical. Our findings that AI training workloads exhibit substantial power fluctuations, particularly in transformer architectures, suggests these facilities may place novel demands on electrical infrastructure. While the industry's economic incentives for efficiency and flexibility in facility siting provide some environmental advantages, the seemingly inelastic demand for AI compute could quickly erode efficiency gains. This underscores the importance of high-resolution empirical data collection and transparency from operators to enable both facility optimization and effective grid integration.

These results point to several key priorities for future research: detailed component-level power measurement to better understand subsystem interactions, systematic investigation of energy implications across model architectures and scales, and development of more sophisticated inference energy models that can account for heterogeneous hardware configurations and varying latency requirements. As AI compute demands continue to grow, such empirically-grounded understanding will be essential for managing energy system impacts while ensuring reliable infrastructure operation.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: <https://doi.org/10.1184/R1/29067572.v1> [110].

Acknowledgment

This manuscript has been authored by an employee of the Lawrence Berkeley National Laboratory under Contract No. DE-AC02-05CH11231 with the U.S. Department of Energy for the Industrial Efficiency and Decarbonization Office (IEDO). The U.S. Government retains, and the publisher, by accepting the article for publication, acknowledges, that the U.S. Government retains a non-exclusive, paid-up, irrevocable, world-wide license to publish or reproduce the published form of this manuscript, or allow others

to do so, for U.S. Government purposes. Thank you to Sarah Smith, Alex Hubbard, Nichole Hanus, Dale Sartor, and Sylvia Downing.

Conflict of interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used *Claude* to improve language, clarity, and type-setting. After using this tool or service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Author contributions

Alex C Newkirk  0000-0002-6213-6865

Conceptualization (equal), Formal analysis (equal), Methodology (equal), Visualization (equal), Writing – original draft (equal), Writing – review & editing (equal)

Jared Fernandez

Conceptualization (equal), Methodology (equal), Writing – original draft (equal), Writing – review & editing (equal)

Jonathan Koomey

Conceptualization (equal), Writing – original draft (equal), Writing – review & editing (equal)

Imran Latif

Investigation (equal)

Emma Strubell

Conceptualization (equal), Resources (equal), Writing – review & editing (equal)

Arman Shehabi  0000-0002-1735-6973

Funding acquisition (equal), Writing – review & editing (equal)

Constantine Samaras  0000-0002-8803-2845

Conceptualization (equal), Supervision (equal), Writing – original draft (equal), Writing – review & editing (equal)

Appendix A. AI training: IT hardware and facility overview

AI applications possess unique attributes that place novel demands on hardware infrastructure compared to previous enterprise computing workloads. State-of-the-art deep learning systems are highly parallelized, benefit tremendously from scale, and require specialized IT hardware. AI chips include graphical processing units (GPUs), field programmable gate arrays (FPGAs), tensor processing units, and some application-specific-integrated-circuits (ASICs) specifically designed for AI workloads. In computer hardware there is a typical trade-off between generality and efficiency/speed; AI applications accentuate these tradeoffs due to the quantity of required operations. AI accelerators perform multiple computations in parallel, and conduct calculations at reduced precision without sacrificing output accuracy [8].

The most general-purpose computer hardware is the central processing unit (CPU), the workhorse of conventional enterprise computer infrastructure. While theoretically capable of the repeated predictable computations required to train and inference AI models, CPUs are at a profound performance disadvantage. Compared to CPUs, AI accelerators are $\sim 10\times$ – $1000\times$ faster and more energy efficient in both training and inferencing applications [8]. These performance gains are so acute that training cutting-edge models relies on the most advanced hardware. Improvements in algorithmic efficiency, computational hardware, and system optimization have enabled higher parameter count and inference accuracy rather than energy performance [46]. Early configurations of AI training hardware were as humble as a single accelerator, by late 2024 the largest clusters contained tens of thousands of individual GPUs [89].

The most common type of AI processor is the GPU, which were originally developed to process computer graphics¹³. In the early 2010s, machine learning researchers found that they could train models faster on GPU integrated hardware [111], and by 2017 GPUs were the hardware architecture in the majority of machine learning applications [112]. GPUs are predominantly used for AI training; they are deployed to a lesser extent for inferencing. While capable of performing robust parallel computation, GPUs are general purpose devices suitable for a variety of applications. TPUs are a related type of processor designed by Google for their internal AI and ML applications. Though technically an ASIC, TPUs are closest in architecture and performance to a machine-learning optimized GPU. Like GPUs, TPUs are best suited to training workloads, though the hardware is sometimes shifted to inferencing when it would otherwise be idle or during peak inference demand [113]. The most popular and highest performance GPUs for AI training are designed by Nvidia, manufactured by Taiwan Semiconductor Manufacturing Company, such as the H100 or B200. For the purposes of our energy use model, we selected Nvidia H100 GPUs as the representative AI training chip, owing to their dominant market share at time of writing. Note the extreme up-front capital cost to building out dedicated AI training infrastructure with these devices, which cost on the order of ~\$40 000 per GPU the year of their release [114].

To optimize model performance, training will iteratively pass data backwards and forwards through neural layers, adjusting weighting according to an optimization function. Additionally, individual calculations will often require input from long-range dependencies or another output, and will thus require interfacing with other neurons, layers, or values stored in memory. In practice this means that during training, an individual parameter, batch, chip, or even rack can be throttled by memory, networking, or just a slower computation elsewhere in the network. Training optimization then batches data and computations to minimize this throttling.

We can build out a description of AI training facilities based on these characteristics. The computation will take place on specialized, accelerated processors optimized for parallel operations and back propagation. As any individual computation often requires input data from another computation or memory, these processors rely on large amounts of memory and high-bandwidth interconnect [115]. To avoid significant throttling, these processors should ideally be in close physical proximity. The continued performance improvement with increased computational load [28, 29, 116] means these will be some of the largest compute scale facilities in the industry. The energy performance of a given training workload is highly sensitive to choice of accelerator. Wang *et al* empirically measured the energy use to train and fine tune the AI model BERT across two different 4-GPU nodes: one with A100s, and one with RTX 8000s [73]. This is a notable comparison as both nodes have the same manufacturer rated TDP of 1500 W, but while the A100 is optimized for AI workloads, the RTX is a more general purpose chip appropriate for a variety of parallel applications such as graphical rendering. Architectural optimization in the A100 resulted in dramatic efficiency gains: training BERT on the A100 Node consumed 146 kWh, while the RTX based node required 368 kWh.

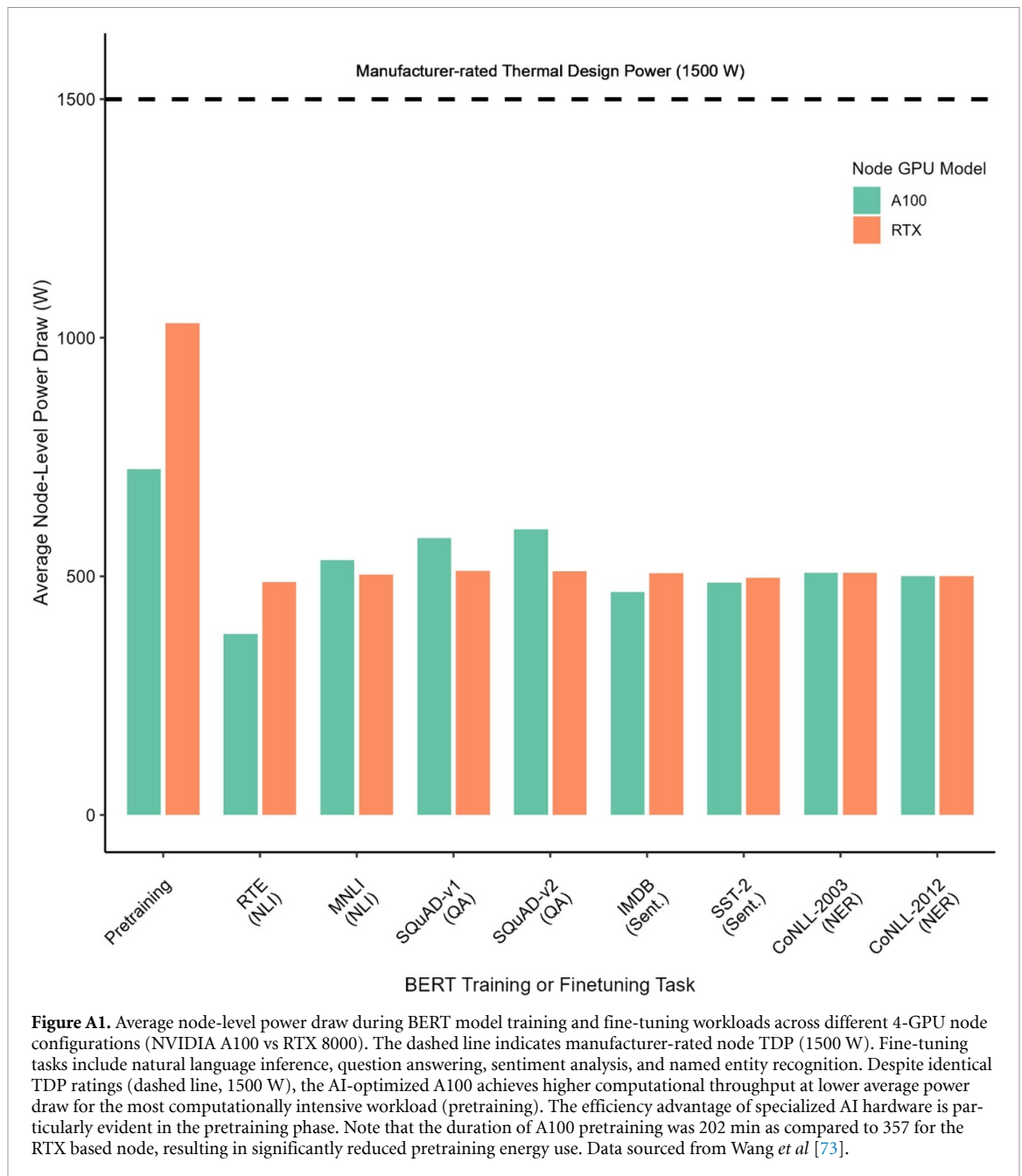
In comparisons between fixed workloads with constant global batch sizes in our data (i.e., Llama-70 B 1 vs 8, and Resnet 1 vs 8), we find that increasing the allocation of computing devices (nodes) increases the total IT energy of the workload—increasing by 60.17% and 148.34% for the Llama-70 B and ResNet training jobs, respectively. Although more devices increases the speed at which a workload is executed, per-node power draw does not decrease enough to offset the energy cost of additional devices. Practitioners therefore face a three-way tradeoff between training speed, energy consumption, and infrastructure costs that should inform their hardware provisioning decisions [70].

Appendix B. Inference energy estimation

In contrast with training, inference energy consumption has only recently garnered significant scholarly attention [26]. A review of 98 papers modeling AI energy use between 2015 and 2022 found only 17 specifically modeled a distinct inferencing energy footprint [117], though inference computational efficiency has long been a topic of interest among ML researchers. Training has a distinct hardware profile and discrete periods of operation, making it conceptually straightforward¹⁴ to estimate energy footprint

¹³ Note that graphical processing, physical simulations, and machine learning are computationally similar. This is not coincidence; graphical processing can be conceptualized as a highly simplified physical model of materials, fluids, and optical behavior.

¹⁴ Note that cloud service providers vary in the level of hardware transparency they afford users, and cloud operators have limited visibility into the algorithms being run on their systems. This informational siloing limits opportunity for hardware-software cooptimization.



[45]. In contrast, inference is a much less distinct load, with more ambiguous hardware. Historic estimates of the inferencing share of ML related energy use for a given AI model deployments range from 35% [35] to 60% [34].

Note that these comparisons of training and inference energy are for historical data before widespread consumer adoption of LLMs. While inference of ML models has been a source of revenue for a decade, these algorithmic ad-targeting and content-recommendation workloads were comparatively modest and computationally streamlined compared to frontier models. Forward looking analysis projects an increasing inference energy share as models become more capable. Median case projection by Schneider Electric estimated inference to draw an 85% share of AI related compute load by 2028 [118].

A major challenge in modeling inference energy use is the heterogeneity of inference hardware. Depending on the output and DNN architecture, ASICs [119], FPGAs [120], GPUs [121], or even CPUs [26, 122] can achieve the highest performance. This then makes modeling of widespread inferencing a challenge, as outside the laboratory inference occurs on a mixed hardware stack. Some research adopts a top-down approach, assuming a given hardware mix, and multiplying number of inference tokens by an efficiency derived from TDP [121] or parameterizing an inference power draw to a fraction of TDP

[93, 99]. Modeling LLM inference demand to match search engine request rate, Chien and coauthors estimated inference energy use to be 1386 times the training cost of GPT-3 [99].

Empirical measurements of hardware power draw are more common in the inference related research than in training [24, 26, 123]. This in part stems from differences in the upfront capital cost of inferring hardware infrastructure as compared to training. While an individual training node costs on the order of hundreds of thousands of U.S. dollars, inference can be performed in a laboratory setting on commercially available hardware. Luccioni, Jernite, and Strubell used empirical inference performance to estimate the required total queries to match training energy use of differently sized models [24]. They estimated query totals on the order of hundreds of millions to match training costs, with widespread consumer adoption such query totals would occur in a few months. Energy performance of inferring can also be improved through architectural and algorithmic optimization [27], and carbon aware scheduling [99]. The inference application modeled by Chien and coauthors is consumer facing generating text, a distinctive workload with a unique latency sensitivity. Anecdotal reports of LLM chatbot product development reference introducing additional delay in response display to better match consumer preferences. Other inference loads such as robotics, algorithmic finance, or vehicle automation may be more latency sensitive and thus possess different load shifting feasibility.

Appendix C. Alternative model specifications and component power behavior

While our chosen model specification maps well onto our quantity of concern, there may be superior alternatives for physically describing the underlying computational system. One such alternative would be a more robust decomposition of the component contributions to node and system power draw to the component level. These cannot be determined from node level data, but we present our hypotheses for the functional forms relating power draw of each to computational demand below:

- Logic components (G, L): Logic component power consumption follows an asymptotic relationship due to architectural limits in parallelization and fundamental voltage-frequency constraints. At low utilization, power consumption scales roughly linearly with frequency while voltage remains at minimum operational levels, reflecting the sum of constant leakage power and dynamic switching losses. However, as computational demands increase beyond what can be achieved at minimum voltage, higher operating frequencies require increased supply voltages, leading to cubic scaling in power consumption due to the combined effect on both switching and leakage losses. This creates a distinct inflection point in the power-performance curve where the scaling behavior transitions from linear to cubic. GPUs display particularly complex behavior, showing non-linear decreases in power consumption with reduction in arithmetic intensity, suggesting an exotic functional form [51].
- Fans (F): A bank of n single speed fans draws power according to stepwise scaling with total heat generation until all n fans are in use, which represents a ceiling of power demand. Alternatively, if each fan is spun by a driven motor capable of variable speed operation, there will be a more complicated scaling as each individual fan draws power depending on the cube of its speed [124, 125]. Fan controllers could then optimize the energy performance of the fan bank depending on the scale and distribution of the thermal load. Precisely characterizing these dependencies, and the energy performance impact from fully shifting thermal management of AI training servers to more efficient technologies, is an opportunity for future empirical work.
- Memory and storage systems (S): High-bandwidth memory power demand increases step-wise linear as memory banks are activated, up to maximum capacity. Storage drive energy use during batch loading and unloading should linearly scale with model size up to a maximum. Standalone metering of the power draw of integrated memory components (as opposed to storage) is an open technical problem, and research should prioritize a non-invasive solution which could generalize across hardware configurations.
- Interconnect (I): power consumption likely follows a piecewise linear relationship with data movement demands, characterized by distinct behaviors at different scales. Within nodes, high-speed GPU-to-GPU links (like NVLink) should show approximately linear power scaling with bandwidth utilization until hitting physical channel limits. For multi-node training, the relationship becomes more complex as power consumption depends on both bandwidth utilization and network topology. For multi-node training, both intra-node (between GPUs) and inter-node communication should be metered to better capture total system energy use.

Appendix D. LOOCV coefficient, workload hyperparameter, and validation set summary tables

Table D1. FLOP calculation summary by workload.

Workload	Architecture	Batch size	Hardware	Flop calculation method	Key Parameters
bnl-resnet 1 [1]	CNN (ResNet152)	512 (64/GPU)	1 node × 8 H100	11.3 GFLOPs/image × 512 × 3	Image size: 32 × 32, Dataset: CIFAR-10 (50 K), Epochs: 200, Learning rate: 0.05
bnl-resnet 2 [1]	CNN (ResNet152)	4096 (512/GPU)	1 node × 8 H100	11.3 GFLOPs/image × 4096 × 3	Image size: 32 × 32, Dataset: CIFAR-10 (50 K), Epochs: 200, Learning rate: 0.05
smc-resnet 2 [1]	CNN (ResNet50)	3200 (400/GPU)	1 node × 8 H100	3.6 GFLOPs/image × 3200 × 3	Image size: 224 × 224, Dataset: ImageNet (1.28 M), Epochs: 35, Learning rate: 0.011
smc-resnet 1 [8]	CNN (ResNet50)	3200 (50/GPU)	8 nodes × 8 H100	3.6 GFLOPs/image × 3200 × 3	Image size: 224 × 224, Dataset: ImageNet (1.28 M), Epochs: 35, Learning rate: 0.011
bnl-llama-13b [1]	LLM (Llama2-13 B)	64	1 node × 8 H100	Narayanan formula(equation (10))	Hidden size: 5120, Layers: 40, Sequence length: 4096, TP: 4, PP: 1, CP: 1, Learning rate: 0.00002
smc-llama-70b [1]	LLM (Llama2-70 B)	4	1 node × 8 H100	Narayanan formula(equation (10))	Hidden size: 8192, Layers: 80, Sequence length: 4096, TP: 4, PP: 1, CP: 1, Learning rate: 0.0004
smc-llama-70b [8]	LLM (Llama2-70 B)	8	8 nodes × 8 H100	Narayanan formula(equation (10))	Hidden size: 8192, Layers: 80, Sequence length: 4096, TP: 4, PP: 1, CP: 2, Learning rate: 0.00036
smc-llama-70b [64]	LLM (Llama2-70 B)	64	64 nodes × 8 H100	Narayanan formula(equation (10))	Hidden size: 8192, Layers: 80, Sequence length: 4096, TP: 4, PP: 1, CP: 2, Learning rate: 0.0005
smc-gpt3-175b [64]	LLM (GPT3-175 B)	128	64 nodes × 8 H100	Narayanan formula(equation (10))	Hidden size: 12 288, Layers: 96, Sequence length: 2048, TP: 4, PP: 8, CP: 1, Learning rate: 0.00002

Table D2. Full LOOCV steepness coefficient stability—potential outlier bolded.

LOOCV results	Asymptotic model	Sigmoid model	
Held-out workload	α	k	x_0
Full data	5.55	1.12	11.46
SMC-GPT3-175 B [64]	5.98	1.28	11.45
SMC-Llama-70 B [64]	5.2	1.07	11.58
SMC-Llama-70 B [8]	6.1	1.18	11.41
SMC-Llama-70 B [1]	5.58	0.97	11.46
BNL-Llama-13 B [1]	6.05	1.31	11.45
SMC-ResNet 2 [1]	5.7	1.11	11.45
SMC-ResNet 1 [8]	5.54	0.83	11.44
BNL-Resnet 1 [1]	4.64	0.006	9.21
BNL—Resnet 2 [1]	5.23	1.24	11.45
LOOCV summary	α	k	x_0
Mean	5.56	1.01	11.24
StdDev	0.43	0.4	0.72
CoV (%)	7.7	39.6	6.4

Table D3. Validation dataset summary.

Workload	Arch.	Parameters	Nodes	P_{avg} (kW)	Energy(kWh)	Duration(Hours)	FLOPS/Node	Source
Dell Llama-70 B [8]	LLM	70 B	8	7.29	5.54	0.10	2.29×10^{15}	Dell
SMC Llama-70 B [8]	LLM	70 B	8	7.73	5.43	0.09	2.29×10^{15}	SMC
UNet-19 M-1 [1]	CNN	19 M	1	6.41	1.43	0.22	4.87×10^{12}	SMC
UNet-19 M-2 [9]	CNN	19 M	9	5.90	3.06	0.06	6.96×10^{11}	SMC

References

- [1] IEA 2024 World energy outlook international energy agency (World Energy Outlook) (available at: www.iea.org/reports/world-energy-outlook-2024)
- [2] LeCun Y, Bengio Y and Hinton G 2015 Deep learning *Nature* **521** 436–44
- [3] Vaswani A *et al* 2017 Attention is all you need *Proc. of the 31st Int. Conf. on Neural Information Processing Systems (Red Hook, NY, USA)* pp 6000–10
- [4] Sutton R 2019 The bitter lesson *Incomplete Ideas* **13** 38 (available at: www.incompleteideas.net/IncIdeas/BitterLesson.html)
- [5] Buchanan B 2020 The AI triad and what it means for national security strategy (Center for Security and Emerging Technology) (available at: <https://cset.georgetown.edu/publication/the-ai-triad-and-what-it-means-for-national-security-strategy/>) (Assessed 7 June 2024)
- [6] Ren J and Xu L 2015 On vectorization of deep convolutional neural networks for vision tasks *Proc. AAAI Conf. on Artificial Intelligence* vol 29 (available at: <https://ojs.aaai.org/index.php/AAAI/article/view/9488>)
- [7] Weng L and Brockman G 2022 Techniques for training large neural networks (available at: <https://openai.com/research/techniques-for-training-large-neural-networks>) (Assessed 20 February 2024)
- [8] Khan S M and Mann A 2020 AI chips: what they are and why they matter (Center for Security and Emerging Technology) pp 22–23 (available at: <https://cset.georgetown.edu/publication/ai-chips-what-they-are-and-why-they-matter/>) (Accessed 12 January 2024)
- [9] Narayanan D *et al* 2021 Efficient large-scale language model training on GPU clusters using megatron-LM *Proc. Int. Conf. for High Performance Computing, Networking, Storage and Analysis (ACM)* pp 1–15 (available at: <https://dl.acm.org/doi/10.1145/3458817.3476209>)
- [10] Masanet E, Shehabi A and Koomey J G 2013 Characteristics of low-carbon data centres *Nat. Clim. Change* **3** 627–30
- [11] Wu C J *et al* 2022 Sustainable AI: environmental implications, challenges and opportunities *Proc. Machine Learning and Systems* pp 795–813
- [12] Kaack L H, Donti P L, Strubell E, Kamiya G, Creutzig F and Rolnick D 2022 Aligning artificial intelligence with climate change mitigation *Nat. Clim. Change* **12** 518–27
- [13] Gupta U *et al* 2020 chasing carbon: the elusive environmental footprint of computing (arXiv:2011.02839)
- [14] Gupta U *et al* 2022 ACT: designing sustainable computer systems with an architectural carbon modeling tool *Proc. 49th Annual Int. Symp. on Computer Architecture (Association for Computing Machinery)* pp 784–99
- [15] Malmodin J, Lövehagen N, Bergmark P and Lundén D 2024 ICT sector electricity consumption and greenhouse gas emissions—2020 outcome *SSRN Electron. J.* **48** 102701
- [16] Acun B *et al* 2023 Carbon explorer: a holistic framework for designing carbon aware datacenters *Proc. 28th ACM Int. Conf. on Architectural Support for Programming Languages and Operating Systems (ASPLOS 2023)* (Association for Computing Machinery) vol 2 pp 118–32
- [17] Wang J *et al* 2024 Designing cloud servers for lower carbon 2024 *ACM/IEEE 51st Annual Int. Symp. Computer Architecture (ISCA)* (IEEE) pp 452–70 (available at: <https://ieeexplore.ieee.org/document/10609689/>)
- [18] Belkhir L and Elmeli A 2018 Assessing ICT global emissions footprint: trends to 2040 & recommendations *J. Clean. Prod.* **177** 448–63

- [19] Schneider I et al 2025 Life-cycle emissions of AI hardware: a cradle-to-grave approach and generational trends (arXiv:2502.01671 [cs.AR])
- [20] Falk S et al 2025 More than carbon: cradle-to-grave environmental impacts of GenAI training on the Nvidia A100 GPU (arXiv:2509.00093 [cs.CY])
- [21] Elsworth C et al Measuring the environmental impact of delivering AI at Google Scale (available at: <http://arxiv.org/abs/2508.15734>) (Assessed 10 November 2025)
- [22] Mytton D 2021 Data centre water consumption *npj Clean Water* **4** 11
- [23] Siddik M A B, Shehabi A and Marston L 2021 The environmental footprint of data centers in the United States *Environ. Res. Lett.* **16** 064017
- [24] Luccioni A S, Jernite Y and Strubell E 2024 Power hungry processing: watts driving the cost of AI deployment? *Proc. 2024 ACM Conf. on Fairness, Accountability, and Transparency* (Association for Computing Machinery) pp 85–99 (available at: <https://dl.acm.org/doi/10.1145/3630106.3658542>)
- [25] Bouza L, Bugeau A and Lannelongue L 2023 How to estimate carbon footprint when training deep learning models? A guide and review *Environ. Res. Commun.* **5** 115014
- [26] Caspart R et al 2022 Precise energy consumption measurements of heterogeneous artificial intelligence workloads *High Performance Computing ISC High Performance 2022 International Workshops* ed H Anzt, A Bienz, P Luszczek and M Baboulin (Springer) pp 108–21
- [27] Del Rey S, Martínez-Fernández S, Cruz L and Franch X 2023 Do DL models and training environments have an impact on energy consumption? *2023 49th Euromicro Conf. on Software Engineering and Advanced Applications (SEAA)* pp 150–8 (available at: <https://ieeexplore.ieee.org/abstract/document/10371661>)
- [28] Kaplan J et al Scaling laws for neural language models (arXiv:2001.08361)
- [29] Brown T B et al 2020 Language models are few-shot learners (arXiv:2005.14165)
- [30] Strubell E, Ganesh A and McCallum A 2019 Energy and policy considerations for deep learning in NLP (arXiv:1906.02243)
- [31] Dodge J et al 2022 Measuring the carbon intensity of AI in cloud instances *Proc. 2022 ACM Conf. on Fairness, Accountability, and Transparency FAccT'22* (Association for Computing Machinery) pp 1877–94
- [32] Patel D, Nishball D and Ontiveros J E 2024 AI datacenter energy dilemma—race for AI datacenter space (available at: www.semianalysis.com/p/ai-datacenter-energy-dilemma-race)
- [33] Patel P et al 2024 Characterizing power management opportunities for LLMs in the cloud *Proc. 29th ACM Int. Conf. on Architectural Support for Programming Languages and Operating Systems (ASPLOS'24)* vol 3 (Association for Computing Machinery) pp 207–22
- [34] Patterson D, Gonzalez J, Hölzle U, Le Q, Liang C, Munguia L-M, Rothchild D, So D R, Texier M and Dean J 2022 The carbon footprint of machine learning training will plateau, then shrink (available at: <http://arxiv.org/abs/2204.05149> [cs.LG])
- [35] Luccioni A S, Viguier S and Ligozat A L 2023 Estimating the carbon footprint of BLOOM, a 176B parameter language model *J. Mach. Learn. Res.* **24** 1–15
- [36] de Vries A 2023 The growing energy footprint of artificial intelligence *Joule* **10** 2191–94 (available at: www.sciencedirect.com/science/article/pii/S2542435123003653)
- [37] Faiz A et al 2023 LLMCarbon: modeling the end-to-end carbon footprint of large language models (arXiv:2309.14393)
- [38] Lacoste A, Luccioni A, Schmidt V and Dandres T 2019 Quantifying the carbon emissions of machine learning (available at: <https://doi.org/10.48550/arXiv.1910.09700>)
- [39] Berthelot A, Jay M, Lefevre L and Caron E 2023 Estimating the environmental impact of generative-AI services using an LCA-based methodology *Procedia CIRP* **122** 707–12
- [40] Touvron H et al 2023 Llama 2: open foundation and fine-tuned chat models (arXiv:2307.09288)
- [41] Meta A I Llama 3 Model Card 2024 (available at: https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md) (Assessed 9 April 2025)
- [42] Henderson P, Hu J, Romoff J, Brunskill E, Jurafsky D and Pineau J 2020 Towards the systematic reporting of the energy and carbon footprints of machine learning (arXiv:2002.05651)
- [43] Farrell S et al 2021 MLPerf™ HPC: a holistic benchmark suite for scientific machine learning on HPC systems *2021 IEEE/ACM Workshop on Machine Learning in High Performance Computing Environments (MLHPC)* pp 33–45 (available at: <https://ieeexplore.ieee.org/abstract/document/9653178>)
- [44] Alben J 2024 Computing in the era of generative AI *2024 IEEE Int. Solid-State Circuits Conf. (ISSCC)* pp 26–28 (available at: <https://ieeexplore.ieee.org/abstract/document/10454562>)
- [45] Patterson D et al 2021 Carbon emissions and large neural network training (arXiv:2104.10350)
- [46] Shankar S and Reuther A 2022 Trends in energy estimates for computing in AI/machine learning accelerators, supercomputers, and compute-intensive applications *2022 IEEE High Performance Extreme Computing Conf. (HPEC)* pp 1–8 (available at: https://ieeexplore.ieee.org/abstract/document/9926296?casa_token=5sj8DvqBG7oAAAAA:ScbNdWr1AQ6ODubRKK5P36SZC-3yV_yLmSUW1h2MbkRVZdj87yBuBGjrPp8QSG18ET2eqQbS)
- [47] Cottier B, Rahman R, Fattorini L, Maslej N and Owen D 2024 The rising costs of training frontier AI models (arXiv:2405.21015)
- [48] Latif I, Newkirk A C, Carbone M R, Munir A, Lin Y, Koomey J, Yu X and Dong Z 2025 Single-node power demand during AI training: measurements on an 8-GPU NVIDIA H100 system *IEEE Access* **13** 61740–7
- [49] Grattafiori A et al 2024 The Llama 3 herd of models (arXiv:2407.21783)
- [50] Groeneveld D et al 2024 OLMo: accelerating the science of language models (arXiv:2402.00838)
- [51] Fernandez J et al 2024 hardware scaling trends and diminishing returns in large-scale distributed training (arXiv:2411.13055)
- [52] Tschand A et al 2025 MLPerf power: benchmarking the energy efficiency of machine learning systems from microwatts to megawatts for sustainable AI (arXiv:2410.12032)
- [53] NVIDIA 2024 NVIDIA DGX superpod data center design reference NVIDIA
- [54] Cameron A C and Miller D L 2010 Robust inference with clustered data *Handbook of Empirical Economics and Finance* (Chapman and Hall/CRC) pp 1–28
- [55] Cameron A C and Miller D L 2015 A practitioner's guide to cluster-robust inference *J. Hum. Resour.* **50** 317–72
- [56] Hodak M, Gorkovenko M and Dholakia A 2019 Towards power efficiency in deep learning on data center hardware *2019 IEEE Int. Conf. on Big Data (Big Data)* pp 1814–20 (available at: <https://ieeexplore.ieee.org/document/9005632>)
- [57] Möbius C, Dargie W and Schill A 2014 Power consumption estimation models for processors, virtual machines, and servers *IEEE Trans. Parallel Distrib. Syst.* **25** 1600–14

- [58] Tsirogiannis D, Harizopoulos S and Shah M A 2010 Analyzing the energy efficiency of a database server *Proc. 2010 ACM SIGMOD Int. Conf. on Management of Dat (SIGMOD'10)* (Association for Computing Machinery) pp 231–42
- [59] Molka D, Hackenberg D, Schöne R and Müller M S 2010 Characterizing the energy consumption of data transfers and arithmetic operations on x86–64 processors *Int. Conf. on Green Computing* pp 123–33
- [60] Akl S G 2006 Three counterexamples to dispel the myth of the universal computer *Parallel Process. Lett.* **16** 381–403
- [61] Amodei D, Hernandez D, Sastry G, Clark J, Brockman G and Sutskever I 2018 AI and compute (available at: <https://openai.com/index/ai-and-compute/>)
- [62] Albanie S 2019 Memory consumption and FLOP count estimates for convnets (available at: <https://github.com/albanie/convnet-burden?tab=MIT-1-ov-file>) (Assessed 18 April 2025)
- [63] Gregersen T, Patel P and Choukse E 2025 Input-dependent power usage in GPUs *Proc. SC'24 Workshops of the Int. Conf. on High Performance Computing, Network, Storage, and Analysis (SC-W'24)* (IEEE Press) pp 1872–7
- [64] Du N et al 2022 GLaM: efficient scaling of language models with mixture-of-experts (arXiv:2112.06905)
- [65] Hoffmann J et al 2024 Training compute-optimal large language models *Proc. 36th Int. Conf. on Neural Information Processing Systems (NIPS'22)* (Curran Associates Inc.) (available at: <http://arxiv.org/abs/2203.15556>)
- [66] Rae J W et al 2022 Scaling language models: methods, analysis & insights from training gopher (arXiv:2112.11446)
- [67] Jay M, Ostapenko V, Lefevre L, Trystram D, Orgerie A C and Fichel B 2023 An experimental comparison of software-based power meters: focus on CPU and GPU 2023 *IEEE/ACM 23rd Int. Symp. on Cluster, Cloud and Internet Computing (Ccgird)* pp 106–18
- [68] Severini T A and Wong W H 1992 Profile likelihood and conditionally parametric models *Ann. Stat.* **20** 1768–802
- [69] Delestrac P et al 2024 Analyzing GPU energy consumption in data movement and storage 2024 *IEEE 35th Int. Conf. on Application-specific Systems, Architectures and Processors (ASAP)* (IEEE) pp 143–51 (available at: <https://ieeexplore.ieee.org/document/10631142/>)
- [70] Li B et al 2022 AI-enabling workloads on large-scale GPU-accelerated system: characterization, opportunities, and implications 2022 *IEEE Int. Symp. on High-Performance Computer Architecture (HPCA)* pp 1224–37 (available at: <https://ieeexplore.ieee.org/abstract/document/9773216>)
- [71] He K, Zhang X, Ren S and Sun J 2015 Deep residual learning for image recognition (arXiv:1512.03385)
- [72] Govind A, Bhalachandra S, Zhao Z, Rrapaj E, Austin B and Nam H A 2023 comparing power signatures of HPC workloads: machine learning vs simulation *Proc. SC'23 Workshops of the Int. Conf. on High Performance Computing, Network, Storage, and Analysis (SC-W'23)* (Association for Computing Machinery) pp 1890–3
- [73] Wang X, Na C, Strubell E, Friedler S and Luccioni S 2023 Energy and carbon considerations of fine-tuning BERT (arXiv:2311.10267)
- [74] Arafa Y et al 2020 Verified instruction-level energy consumption measurement for NVIDIA GPUs *Proc. 17th ACM Int. Conf. on Computing Frontiers* pp 60–70 (available at: <http://arxiv.org/abs/2002.07795>)
- [75] You J, Chung J W and Chowdhury M 2023 Zeus: understanding and optimizing {GPU} energy consumption of {DNN} training 2023 *20th USENIX Symp. on Networked Systems Design and Implementation (NSDI 23)* pp 119–39 (available at: www.usenix.org/conference/nsdi23/presentation/you)
- [76] Yang Z, Adamek K and Armour W 2024 Part-time power measurements: nvidia-smi's lack of attention (arXiv:2312.02741)
- [77] Shukla P, Aguren D, Burd T, Coskun A K and Kalamatianos J 2023 Temperature-aware sizing of multi-chip module accelerators for multi-DNN workloads 2023 *Design, Automation & Test in Europe Conf. & Exhibition (DATE)* pp 1–6 (available at: <https://ieeexplore.ieee.org/abstract/document/10137047>)
- [78] Mei X, Yung L S, Zhao K and Chu X 2013 A measurement study of GPU DVFS on energy conservation *Proc. Workshop on Power-Aware Computing and Systems (HotPower'13)* (Association for Computing Machinery) pp 1–5
- [79] Price D C, Clark M A, Barsdell B R, Babich R and Greenhill L J 2016 Optimizing performance-per-watt on GPUs in high performance computing *Comput. Sci. Res. Dev.* **31** 185–93
- [80] Acun F, Zhao Z, Austin B, Coskun A K and Wright N J 2024 Analysis of power consumption and GPU power capping for MILC SC24-W: *Workshops of the Int. Conf. for High Performance Computing, Networking, Storage and Analysis* (IEEE) pp 1856–61 (available at: <https://ieeexplore.ieee.org/document/10820715/>)
- [81] Tang Z, Wang Y, Wang Q and Chu X 2019 The impact of GPU DVFS on the energy and performance of deep learning: an empirical study *Proc. Tenth ACM Int. Conf. on Future Energy Systems (e-Energy'19)* (Association for Computing Machinery) pp 315–25
- [82] Stojkovic J et al 2025 TAPAS: thermal- and power-aware scheduling for LLM inference in cloud platforms (arXiv:2501.02600)
- [83] Kong R, Zhang H, Tang M, Zou H, Tian C and Ding T 2024 Enhancing data center cooling efficiency and ability: a comprehensive review of direct liquid cooling technologies *Energy* **308** 132846
- [84] Beberide D, Regany D, Camarasa J, Barrau J and Villarrubí M 2024 Performance assessment of an advanced direct-to-chip liquid cooling solution in real conditions inside a data centre 2024 *30th Int. Workshop on Thermal Investigations of ICs and Systems (THERMINIC)* pp 1–5 (available at: <https://ieeexplore.ieee.org/abstract/document/10732550>)
- [85] Narayanan A, Wang Q, Ozguc S and Bonner R W 2024 Investigation of server level direct-to-chip two phase cooling solution for high power GPUs *ASME Digital Collect.* (<https://doi.org/10.1115/1PACK2024-141261>)
- [86] Kulkarni D, Tipton R, Leka E, Bankston J, Wang Q and King J 2025 Pumped 2p refrigerant- based direct liquid cooling (DLC) technology for next generation AI clusters with high TDP accelerators Open Compute Project (available at: www.opencompute.org/documents/2p-refrigerant-based-dlc-wp-v1-pdf-1)
- [87] Huang Y, Liu B, Xu S, Bao C, Zhong Y and Zhang C 2024 Experimental study on the immersion liquid cooling performance of high-power data center servers *Energy* **297** 131195
- [88] Latif I, Shafique M A, Ullah H, Newkirk A C, Yu X and Munir A 2025 Cooling matters: benchmarking large language models and vision-language moavailable at: dels on liquid-cooled versus air-cooled H100 GPU systems (arXiv:2507.16781)
- [89] Dave P and Knight W Meta's next llama AI models are training on a GPU cluster 'bigger than anything' *Else. Wired* (available at: www.wired.com/story/meta-llama-ai-gpu-training/)
- [90] Google Data Centers 2024 Efficiency—data centers—Google (available at: www.google.com/about/datacenters/efficiency/) (Assessed 21 November 2024)
- [91] United States Environmental Protection Agency (EPA) 2024 2022 Emissions & generation resource integrated database (eGRID) (Office of Atmospheric Protection, Clean Air Markets Division) (available at: www.epa.gov/egrid)
- [92] Masanet E, Lei N and Koomey J 2024 To better understand AI's growing energy use, analysts need a data revolution *Joule* **8** 2427–36
- [93] Shehabi A et al 2024 United States data center energy usage report (Lawrence Berkeley National Lab)

- [94] Masanet E, Shehabi A, Lei N, Smith S and Koomey J G 2020 recalibrating global data center energy-use estimates *Science* **367** 984–6
- [95] Jones N 2018 How to stop data centres from gobbling up the world's electricity *Nature* **561** 163–6
- [96] Shehabi A, Smith S J, Masanet E and Koomey J G 2018 Data center growth in the United States: decoupling the demand for services from electricity use *Environ. Res. Lett.* **13** 124030
- [97] Zhang Y, Zhao Y, Dai S, Nie B, Ma H, Li J, Miao Q, Jin Y, Tan L and Ding Y 2022 Cooling technologies for data centres and telecommunication base stations—a comprehensive review *J. Clean. Prod.* **334** 130280
- [98] Moss S 2023 Meta details AI data center redesign that led to facilities being scrapped (available at: www.datacenterdynamics.com/en/analysis/meta-details-ai-data-center-redesign-that-led-to-facilities-being-scrapped/)
- [99] Chien A A, Lin L, Nguyen H, Rao V, Sharma T and Wijayawardana R 2023 Reducing the carbon impact of generative AI inference (today and in 2035) *Proc. 2nd Workshop on Sustainable Computer Systems (HotCarbon'23)* (Association for Computing Machinery) pp 1–7
- [100] Executive Order 14110 Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, 88 Fed. Registry
- [101] Li Y, Mughees M, Chen Y and Li Y R 2024 The unseen AI disruptions for power grids: LLM-induced transients (arXiv:2409.11416)
- [102] Bates N, Ghatikar G, Abdulla G, Koenig G A, Bhalachandra S, Sheikhalishahi M, Patki T, Rountree B and Poole S 2015 Electrical grid and supercomputing centers: an investigative analysis of emerging opportunities and challenges *Inform. Spektrum* **38** 111–27
- [103] Choukse E, Warriar B, Heath S, Belmont L, Zhao A, Khan H A and Saini N 2025 Power stabilization for AI training datacenters (arXiv:2508.14318)
- [104] Valverde G, Pierrou G, Krajacic J and Hug G Forced oscillations in power systems induced by AI workloads in data centers *Authorea Prepr.* (<https://doi.org/10.36227/techrxiv.176127304.46362185/v1>)
- [105] Abdulveleev I R, Khramshin T R, Kornilov G P and Abdulveleeva R R 2021 Experimental study of the impact of a DC electric arc furnace on a power grid 2021 *Int. Conf. on Industrial Engineering, Applications and Manufacturing (ICIEAM)* pp 214–8 (available at: <https://ieeexplore.ieee.org/abstract/document/9446320>)
- [106] Yan J, Huang Y, Gupta A, Gupta A, Liu C, Li J and Cheng L 2022 Energy-aware systems for real-time job scheduling in cloud data centers: a deep reinforcement learning approach *Comput. Electr. Eng.* **99** 107688
- [107] Fahad M, Shahid A, Manumachu R R and Lastovetsky A 2019 A comparative study of methods for measurement of energy of computing *Energies* **12** 2204
- [108] Hu K and Hu K 2023 ChatGPT sets record for fastest-growing user base—analyst note *Reuters* (available at: www.reuters.com/technology/chatgpt-sets-record-fastest-growing-user-base-analyst-note-2023-02-01/)
- [109] Patel D and Nishball D. 2024 Nvidia Blackwell Perf TCO Analysis—B100 vs B200 vs GB200NVL72 (available at: www.semianalysis.com/p/nvidia-blackwell-perf-tco-analysis)
- [110] Alex C N 2025 AI Training Node Level Power Draw *Carnegie Mellon University* (<https://doi.org/10.1184/R1/29067572.v1>)
- [111] Miller C 2022 *Chip War: The Fight for the World's Most Critical Technology* (Scribner) p 464
- [112] Ben-Nun T and Hoeftler T 2019 Demystifying parallel and distributed deep learning: an in-depth concurrency analysis *ACM Comput. Surv.* **52** 1–43
- [113] Moorhead P 2024 Google provides more details on its cloud generative AI play at next event *Forbes* (available at: www.forbes.com/sites/patrickmoorhead/2023/09/12/google-provides-more-details-on-its-cloud-generative-ai-play-at-next-event/)
- [114] Nvidia C D 2024 Powered by A.I. Boom Reports Soaring Revenue and Profits *The New York Times* (available at: www.nytimes.com/2024/05/22/technology/nvidia-quarterly-earnings-results.html)
- [115] Ma R, Georganas E, Heinecke A, Gribok S, Boutros A and Nurvitadhi E 2022 FPGA-based AI smart NICs for scalable distributed AI training systems *IEEE Comput. Arch. Lett.* **21** 49–52
- [116] Sartor S and Thompson N Neural scaling laws for embodied AI
- [117] Verdecchia R, Sallou J and Cruz L 2023 A systematic review of green AI *WIREs Data. Mining. Knowl. Discov.* **13** e1507
- [118] Avelar V and Donovan P 2023 *The AI Disruption: Challenges and Guidance for Data Center Design* (Schneider Electric)
- [119] Jouppi N P et al 2017 In-datacenter performance analysis of a tensor processing unit *Proc. 44th Annual Int. Symp. on Computer Architecture (ISCA'17)* Association for Computing Machinery pp 1–12
- [120] Qasaimieh M, Denolf K, Lo J, Vissers K, Zambreno J and Jones P H 2019 Comparing energy efficiency of CPU, GPU and FPGA implementations for vision kernels 2019 *IEEE Int. Conf. on Embedded Software and Systems (ICES)* pp 1–8 (available at: <https://ieeexplore.ieee.org/abstract/document/8782524>)
- [121] Desislavov R, Martínez-Plumed F and Hernández-Orallo J 2023 Trends in AI inference energy consumption: beyond the performance-vs-parameter laws of deep learning *Sustain. Comput.* **38** 100857
- [122] Hazelwood K et al 2018 Applied machine learning at Facebook: a datacenter infrastructure perspective 2018 *IEEE Int. Symp. on High Performance Computer Architecture (HPCA)* pp 620–9 (available at: <https://ieeexplore.ieee.org/document/8327042>)
- [123] Samsi S et al 2023 From words to watts: benchmarking the energy costs of large language model inference 2023 *IEEE High Performance Extreme Computing Conf. (HPEC)* pp 1–9 (available at: <https://ieeexplore.ieee.org/abstract/document/10363447>)
- [124] Wang Z, Bash C, Tolia N, Marwah M, Zhu X and Ranganathan P 2009 Optimal fan speed control for thermal management of servers *ASME 2009 InterPACK Conf. (ASME/EDC)* vol 2 pp 709–19 (available at: <https://asmedigitalcollection.asme.org/InterPACK/proceedings/InterPACK2009/43604/709/335758>)
- [125] Jorgensen R 1999 *Fan Engineering* 9th edn